



Tax Policy in Connection with Economic Growth: Ghana's Perspective

Master's Thesis

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ABSTRACT

This thesis explores the relatedness between taxation and Ghana's economic development. Utilizing time series data from 1990 to 2022, the study employed the Johansen test, the Granger causality test, and the vector error correction model for analysis. The cointegration results revealed a long-term interdependence among the variables under consideration. The analysis suggested a positive and sustained impact of capital expenditure on Ghana's economic advancement, while a negative relationship was observed between revenue expenditure and growth. The Granger causality test supported a unidirectional link from tax revenue to economic development. Based on these findings, the study recommends that the government broaden the tax base by incorporating more economic activities and sectors. Furthermore, enhancing tax administration effectiveness is advised to bolster the country's economic growth efforts.

Keywords:

Gross Domestic Product, Total Tax Revenue, Capital Expenditure, Revenue Expenditure, Vector Error Correction Model.

ABSTRAKT

Tato práce zkoumá souvislost mezi zdaněním a hospodářským rozvojem Ghany. Studie využívá časové řady dat z let 1990 až 2022 a k analýze používá Johansenův test, Grangerův test kauzality a vektorový model korekce chyb. Výsledky kointegrace odhalily dlouhodobou vzájemnou závislost mezi zkoumanými proměnnými. Analýza naznačila pozitivní a trvalý vliv kapitálových výdajů na hospodářský pokrok Ghany, zatímco mezi výdaji na příjmy a růstem byl pozorován negativní vztah. Grangerův test kauzality potvrdil jednosměrnou vazbu mezi daňovými příjmy a hospodářským rozvojem. Na základě těchto zjištění studie doporučuje, aby vláda rozšířila daňovou základnu zahrnutím více ekonomických činností a odvětví. Dále se doporučuje zvýšit efektivitu daňové správy, aby se posílilo úsilí o hospodářský růst země.

Klíčová slova:

Hrubý Domácí Produkt, Celkové Daňové Příjmy, Kapitálové Výdaje, Příjmové Výdaje, Vektorový Model Korekce Chyb.

Table of Contents

ACKNOWLEDGEMENT.....	<i>i</i>
STATUTORY DECLARATION.....	<i>ii</i>
ABSTRACT.....	<i>iii</i>
1. INTRODUCTION.....	1
1.1 Problem Statement	3
1.2 Objective of the Study	5
1.3 Research Hypothesis	5
1.4 Significance of the Study	5
2. LITERATURE REVIEW.....	6
2.1 The Concept of Taxation.....	6
2.2 Tax Policy in Ghana.....	7
2.3. The Ghanaian Tax System.....	8
2.3.1 Direct Taxes.....	9
2.3.2 Indirect Taxes.....	12
2.3.3 International trade taxes in Ghana	14
2.4. Reforms in the tax system in Ghana	14
2.4.1 Broadening the Tax Base	17
2.4.3 Improving Tax Efficiency and Equity.....	18
2.5. Challenges of the Tax Reforms	20
2.6. Tax Revenue Composition in Ghana	20
2.7. Economic Growth	21
2.7.1 Economic Growth and Government Expenditure.....	23
2.8. Inflation and Economic growth.....	25
2.9. Theoretical Review	25
2.9.1 Exogenous Growth Theories.....	25
2.9.2 Theories of Endogenous of Growth	28
2.9.3 Empirical Review	29
3. METHODOLOGY.....	35
3.1 Overview of Methodology	35
3.2 Data Collection and Variables	36
3.3 Estimation Technique	37
3.4 Model Specification	38
4. DATA ANALYSIS AND RESULTS	42
4.1 Summary Statistics	42
4.2 Collinearity Diagnostic	44
4.3 Vector Autoregression (VAR) Lag-Length Selection.....	45
4.4 Stationarity Test.....	46

4.5 Johansen Cointegration Test	48
4.6 Vector Error Correction Model.....	49
4.7 Granger Causality Test.....	55
4.8 Diagnostic Test	56
4.9 Summary of Results.....	57
5. <i>DISCUSSIONS</i>	58
5.1 What causal connection exists between Ghana's economic growth and its taxation system?	58
5.2 How does Ghana's economic growth relate to Capital Expenditures?	59
5.3 Does Revenue Expenditure influence economic growth in Ghana?	60
6. <i>CONCLUSION AND RECOMMENDATIONS</i>	61
6.1 Conclusion	61
6.2 Recommendations.....	62
6.3 Limitation of this study	62
6.4 Future Research.....	63
7. <i>REFERENCES:</i>	64
8. <i>List of Figures</i>	72
9. <i>List of Tables</i>	73

1. INTRODUCTION

Economic growth remains a crucial objective for all countries regardless of their development status. It refers to the increase in an economy's capacity to generate products and services when comparing different time periods. The duty of providing residents' access to necessary infrastructure rests on the governments. Income redistribution, economic stabilisation, and the provision of economic services are among the main duties that governments have toward their constituents. The economy of Ghana has flourished tremendously in recent years. Over the course of ten years leading up to 2020, the country's real GDP expanded by an impressive 92 percent (Bank of Ghana, 2021). Apart from the macroeconomic progress, the economy has experienced notable systemic transformations. The government's ability to fulfil its responsibilities largely depend on the revenue generated from various sources, both internal and external. Taxation is one of the crucial sources of revenue. Taxation has a long-standing history as a means to fund government expenditures. It serves as a tool for enhancing public sector performance and repaying debts.

Taxation plays a critical role in any country's pursuit of self-sufficiency and compliance with economic regulations. The purpose of taxation is to enable governments to achieve their financial goals. By encouraging or discouraging particular purchases, it might affect how consumers behave. It can also be utilized to stimulate or deter investment in specific industries, aiming to regulate economic activities that are deemed harmful or antisocial but not necessarily illegal. Additionally, taxation can be employed to protect local and small businesses and enable them to compete more effectively against larger international competitors. The relationship between tax revenue and economic growth has sparked extensive debate among policymakers, researchers, academics, and economists in developing and emerging economies. Regardless of the direction of influence, the primary concern for most policymakers revolves around the factors that drive increased economic growth. The significance of a government's role in allocating tax revenue for financing public expenditure cannot be overstated. This role serves as a vital tool for achieving objectives such as price stabilization, fostering economic growth, and promoting overall development. Given the increasing public expenditures in both developing and affluent nations, generating sufficient revenue is essential for all governments. Mobilizing adequate resources through taxation becomes imperative to meet expenditure requirements in areas such as road infrastructure, education facilities, healthcare services, defence, and other fundamental amenities is crucial for the well-being of citizens.

Taxation has both microeconomic and macroeconomic implications. On a microeconomic level, it influences income distribution among individuals and promotes resource efficiency. At the macroeconomic level, taxation impacts factors such as capacity output, employment levels, pricing, and overall economic growth (Amanja, 2005). In achieving economic growth, the government plays a crucial role, regardless of whether it is in an established or emerging economy (Acosta-Ormaechea, 2018 and Ahmad, 2016). Fiscal management, including taxation emerges a paramount instrument for the government to nurture economic expansion. Taxation as a fundamental component of fiscal policy, holds significant importance as it directly impacts a country's growth trajectory. It provides a stable and predictable fiscal framework for emerging economies, enabling them to foster growth and finance their social and physical infrastructure requirements (Ricardo, 1951).

The tax structure of a country, as highlighted by Garba (2014), is a key driver of other macroeconomic variables. For instance, in the case of Ghana, tax revenue plays a crucial role as a source of government spending for development (Takumah, 2014). It underpins the country's economic activities and supports its overall developmental goals. Taxation holds significant strategic importance in emerging countries as it serves as a powerful instrument for driving economic development. Taxation has a crucial role in promoting economic advancement by supporting the supply of public goods including infrastructure, healthcare, education, and justice. The impact of taxation extends to various aspects, including individual savings, employment decisions, education choices, production levels, job creation, investment patterns, corporate innovation, and the preferences of investors regarding savings instruments and assets (OECD, 2019). These choices are shaped not solely by tax rates but also by the formulation and incorporation of diverse fiscal instruments utilized to generate government revenue. The manner in which taxes are financed and administered is directly related to the economic progress of a country, which makes them essential components of any effort to achieve long-term development.

Numerous approaches were used in the research, such as the Granger causality test, vector autoregressive lag-selection, stationarity test, Johansen cointegration test, and vector error correction model. The following is the format of the thesis: the initial chapter acts as an introduction, revealing pertinent details about the study's backdrop. It presents the problem statement and outlines the research inquiries that shape the study's objectives. Moreover, it highlights the study's significance. Moving to the second chapter, it establishes an expansive framework for subsequent sections. This chapter extensively evaluates existing research on the concepts and theories that contribute to a comprehensive understanding of the subject matter.

Chapter three delves into the study's methodology, encompassing the sources of data and model specifications. It also incorporates an exhaustive review of the methodology employed. Chapter four is dedicated to the comprehensive analysis and interpretation of results while chapter five deals with in-depth discussions of the findings. In parallel, Chapter six encapsulates conclusions drawn from the study's findings and presents recommendations and prospects for future research on the subject matter.

1.1 Problem Statement

Ghana has experienced a persistent imbalance between government spending and domestic tax collection. Taxation serves as a crucial means for governments to generate the necessary funds to support their expenditures, and it is essential for expenditure to align with revenue collection in order to sustain rapid growth. However, in many developing nations including Ghana, there is a growing disparity between government expenditures and revenue (Twerefou et al., 2010). This disparity is further exacerbated by the need to meet the socioeconomic aspirations of the population, which is crucial for promoting economic growth and development. The disconnect between government spending and revenue collection throws the economy under financial subjugation, leading to chronic budgetary shortfalls. To address this issue, it is vital for the Ghanaian government to implement policies that increase revenue collection. By closing the gap between Ghana's income base and expenditures, the government can alleviate financial pressures and foster a more sustainable economic environment. Taxation plays a crucial role in a country's economic growth, but it can also hinder progress if implemented in an anti-economic manner. There are several ways in which taxation can have adverse effects on growth, such as high taxes, multiple taxation, and double taxation.

High taxation occurs when the government imposes excessive taxes on a company's profits or an individual's earnings. This can reduce the investable funds available to taxpayers, limiting their ability to make productive investments and hindering economic growth. Multiple taxation occurs when different taxes are levied on the same income, leading to a cumulative burden on taxpayers. This can create a complex and burdensome tax system, discouraging business activities and investment. Double taxation arises when the same income is taxed twice as it moves from one geographic area or country to another. This can create barriers to cross-border investments and hinder international trade and economic integration. These forms of taxation reduce the funds available for investment and economic activities, impeding growth and development. On the other hand, if taxpayers evade or avoid paying their fair share of

taxes, it can result in insufficient government revenue to deliver essential services. In such cases, companies may need to allocate more resources to fund operations that would otherwise be supported through tax revenue. Funding social infrastructure and services through a tax revenue pool is generally more sustainable and efficient than relying solely on corporate funding. To foster economic growth, it is important for governments to strike a balance in taxation, ensuring that it is fair, transparent, and supportive of productive economic activities while providing the necessary revenue to fund essential public services.

Tax regulations often lack sufficient adaptation to the unique circumstances of individual taxpayers, and they frequently fail to consider the limited administrative capabilities of the region. Recognizing this challenge, governments in the area have to initiate reforms aimed at alleviating the burden imposed by tax structures that hinder economic progress. These reforms must be designed to promote savings, investment, entrepreneurship, and labour. The objective is to reshaping the tax system to minimize the adverse effects on growth while ensuring adequate fiscal receipts. Although these reforms have sometimes led to lower tariffs and tax rates, they have not always been successful in expanding the domestic tax base. The impact of tax revenue on economic performance, especially in developing countries, has been a subject of empirical dispute (Babatunde et al., 2017; Garba, 2014). Income tax applies to both individual and corporate earnings. However, high taxes on workers' remuneration can discourage work effort and hinder the development of human capital. It may also have the potency of deterring individuals from saving for their future.

Likewise, high taxes on profits can deter investment and discourage entrepreneurship, leading to a decrease in economic production. Conversely, lower taxes can incentivize individuals to work harder, save more, and invest more, thereby increasing overall productivity in the economy. The question of what drives long-term economic growth has been the topic of numerous debates, with arguments stemming from either both demand and supply sides of the economic growth equation. These debates have stimulated further research and exploration into the subject. The imperative to reform Ghana's tax system in order to generate sufficient revenue has long been overdue. Ghana consistently records lower tax revenue, significantly below the recommended 15% tax-to-GDP ratio as advised by the OECD. This ongoing trend raises concerns, as the government often needs to compensate for the inadequate revenue collection by resorting to public debt, which carries negative consequences. Given the challenging fiscal outlook faced by the government, policymakers in Ghana must thoroughly assess the structure of the tax system. Since tax revenues play a crucial role in fostering national wealth creation, the government should prioritize the formulation, administration, and enforcement of effective

tax policies. It's within this context that the Ghanaian government aims to achieve a tax-to-GDP ratio of 20% by 2024, with the aim of supporting public investments and social programs (Ministry of Finance, 2020).

1.2 Objective of the Study

The primary aim of this study is to assess the role of taxation as a catalyst for economic growth in Ghana. In particular, the research seeks to:

1. Assess the association between taxation and economic advancement in Ghana.
2. Explore the linkage between capital expenditure and development in Ghana's economy.
3. Investigate the connection between revenue expenditure and Ghana's economic expansion.

1.3 Research Hypothesis

The summary of the study was drawn from these hypotheses:

- 1: Is there any substantial linkage between taxation and economic advancement in Ghana?
- 2: Is there any significant association between capital expenditure and socio-economic progress in Ghana?
- 3: Is there a notable link between revenue expenditure and economic expansion in Ghana?

1.4 Significance of the Study

Gaining insight into the precise impact of taxation on economic growth is crucial for informed policymaking. Inversely, expounding the connections between these economic indicators is paramount. This implies that government entities can use taxes as economic planning instrument to impact lucrative conditioning, aligning with Keynesian propositions. On the other hand, if no similar relationship is established from taxation to economic expansion, as posited by Classical growth propositions in the long-term, taxation as a financial policy may warrant effectiveness. In spite, the apparent policy applicability of the linkage between taxation and growth, the literature has maintained a reticent station on this issue. This paper trials to address this gap by conducting a thorough examination of the nature of the dynamics between taxation and economic development, specifically in the situation of Ghana.

2. LITERATURE REVIEW

In this chapter, the existing body of literature concerning tax policy and its impact on economic growth is thoroughly examined. The initial section offers a conceptual analysis, delving into the precise definitions of key terms within the realm of taxation and economic policy. The following section is dedicated to a comprehensive theoretical examination where prominent theories regarding the relationship between taxation and economic growth are explored in depth. The subsequent portion undertakes empirical review pertaining to the interplay between taxes and economic growth.

2.1 The Concept of Taxation

Taxation, encompassing a broad spectrum of strategies and instruments, can be perceived as a governmental method to compel monetary funds or valuable assets from both individuals and organizations within society through the enactment of legal frameworks. Governments frequently utilize the tax framework to redistribute monetary resources from affluent entities to those less privileged. Multiple rationales underlie the imposition of taxes by governments, with the foremost being the generation of revenue for public expenditures and, conversely, the efficient allocation of resources. In the context of the national government, taxation stands as its primary source of income (Beza 2014). Predominantly, taxes are imposed by governments to amass funds for the support of public services, while concurrently optimizing the allocation of resources. The crux of government revenue hinges upon taxation (Beza 2014). To accomplish this goal, a progressive tax system that raises average and exemption rates on higher incomes is often used. Taxation serves as a pivotal instrument in the governmental repertoire for augmenting revenue and invigorating economic growth. Progressive, proportional, or regressive tax collection can be achieved through both direct and indirect means.

Operating a vehicle with low or no fuel can be likened to a situation where there are insufficient funds for development. According to Otchere-Ankrah, Senior and Junior (2014), a thriving economy hinges on the contribution of taxes. As elucidated in the preceding paragraphs, the fundamental role of taxation is to amass funds from the general populace and economically active individuals to cover government expenditures. As a matter of routine, citizens are obliged to return a proportionate portion of their income to the state's treasury. In the absence of domestic tax revenue generation, governmental endeavours aimed at fostering development through the provision of essential social amenities and infrastructure may prove

futile (Mbilla et al., 2018). In a developing country like Ghana, merely 20% of the workforce is officially recognized and subjected to taxation. This implies that a substantial 80% of the working population in the country is exempt from tax obligations (Naporow, 2015). Tax adherence carries significant importance in a burgeoning economy such as Sub-Saharan Africa, where self-employed individuals constitute an average of 70% of enterprises (Adom and Williams, 2012). While taxation might be perceived as a hindrance to individuals aspiring for an improved quality of life or to businesses striving for profit generation, the government and its fiscal imperatives regard it as the bedrock and enabler of economic advancement, as indicated by Olusanya et al. (2012). Tax hikes and the integration of novel technologies have acted as catalysts for economic growth and advancement.

2.2 Tax Policy in Ghana

Tax policy encompasses a nation's regulation on the extent and recipients of taxation. It serves several primary objectives, including revenue collection, resource allocation, income redistribution, and the reduction of wealth disparities among citizens. Tax policies can be designed to finance budget deficits or counteract various economic factors, as observed by (Romer and Romer 2010). They enable governments to fulfil essential functions such as providing public goods, maintaining societal order, and ensuring defense against external threats (Edame and Okoi 2014; Takumah 2014). Efforts to enhance tax collection capabilities, mitigate existing distortions, and promote economic incentives have been pursued. Concurrently, efforts have been made to boost the general effectiveness and equitable nature of the tax system. As a result, these tax changes are typically divided into three phases that overlap: rebuilding the tax base, increasing production incentives, and enhancing the efficiency and equality of tax administration. Economic development necessitates active government involvement, both in developed and developing nations, as emphasized by Edame and Okoi (2014). Fiscal policy, therefore, becomes a crucial tool in the government's arsenal for fostering economic growth, with taxation playing a pivotal role. Garba (2014) found that a country's tax structure significantly impacts various macroeconomic indicators. In Ghana, for instance, the government's ability to finance development spending heavily relies on tax revenues, as highlighted by Takumah (2014).

In the Ghanaian context, indirect taxes have historically served as the primary source of revenue. Tax policies aimed at augmenting indirect tax revenue must consider this when determining tax rates. Striking a balance between encouraging imports and generating revenue

is a complex endeavour. Consequently, new tax policies have been introduced, accompanied by adjustments in tax rates for different tax units. Additionally, measures have been implemented to address potential tax loopholes, ensuring equity within the tax system. Despite these efforts, tax administration still encounters challenges. Ghana's tax policy assumes a pivotal role at this juncture in the nation's development, as highlighted (Keene et al., 2019). The significant gap between public expenditures and domestic revenue has led to a substantial dependence on foreign aid. While calls for reducing public spending have been voiced, practical challenges persist. The reality remains that the government must increase investments in critical sectors like healthcare, education, and infrastructure to achieve substantial and sustained economic growth.

2.3. The Ghanaian Tax System

Each nation operates under a tax system overseen by a governmental entity. In the case of Ghana, the Ghana Revenue Authority (GRA) is the legally designated agency responsible for tax administration on behalf of the government. Prior to the formation of the GRA, tax administration was conducted by four distinct entities: the Internal Revenue Service, the Value Added Tax, the Customs, Excise and Preventive Service, and the Revenue Agencies Governing Board. In 2009, these organizations were consolidated to establish the GRA, which is structurally segmented into three primary divisions: the Customs Division, the Domestic Tax Revenue Division, and the Support Service Division.

While the Support Service Division primarily handles administrative functions, the other two divisions are tasked with the administration of various taxes. The formation of the GRA represented a managerial revamp of the tax system. Recently, the GRA has been working on a series of major administrative reforms outlined in three strategic plans. As part of these goals, the Authority was planned to be integrated and thoroughly modernized so that it could carry out its duty; to strengthen enforcement mechanisms; to broaden the tax base; improve processes and procedures; and to execute creative strategies to boost tax revenue collecting. The GRA divides taxes into three categories: income and profit taxes (direct taxes), goods and service taxes (indirect taxes), and international trade taxes (indirect taxes or customs taxes)

Table 1: Categorization of taxes in Ghana

Tax	Tax base
Direct taxes	
Corporate income tax (CIT)	Corporate profits
Personal income tax (PIT)	Individual earnings
Mineral royalties	Revenues from mineral sales
National Fiscal Stabilisation Levy (NFSL)	Pre-tax profit in selected sectors
Stamp duty	Transaction value
Airport tax	Flight passengers
Indirect taxes	
Value added tax	Taxable consumption
National Health Insurance Levy	All taxable sales
Ghana Education Trust Fund Levy	All taxable sales
Petroleum taxes	Petroleum purchases
Import duty	Imported goods
Communication service tax (CST)	Electronic communication services
Excise duty	Selected goods like alcohol, tobacco etc.

Source: Ghana Revenue Authority

2.3.1 Direct Taxes

This tax is imposed directly on individuals or organizations upon which it is applied. This category of taxation encompasses levies on income and property, directly targeting the taxpayers. In Ghana, there are three primary categories of income tax: Personal income tax, corporate income tax, and other forms. Personal income tax comprises Pay-as-you-earn (PAYE) contributions and taxes for self-employed individuals. PAYE deductions involve taking a portion of employees' salaries or wages at the source to fulfil their income tax obligations. The introduction of PAYE aims to alleviate the tax burden on employees, with the tax being deducted automatically from their earnings each pay cycle. These payments stand as the final tax liability, and employees are not required to file additional tax returns unless they hold multiple jobs or possess additional income sources beyond regular employment. The calculation of Pay-as-you-earn (PAYE) relies on the rates of personal income tax (PIT). The current monthly and annual PIT rates are provided in Tables 2 and 3, respectively.

Table 2: PIT Monthly tax rate in Ghana-Effective 1st May, 2023

Chargeable Income (GH¢)	Rate (%)	Tax Payable (GH¢)	Cumulative Income (GH¢)	Cumulative Tax (GH¢)
First 402.00	Free	NIL	402.00	NIL
Next 110.00	5	5.50	512.00	5.50
Next 130.00	10	13.00	642.00	18.50
Next 3,000.00	17.5	525.00	3,642.00	543.50
Next 16,395.00	25	4,098.75	20,037.00	4,642.25
Next 20,000.00	30	8,988.90	50,000.00	13,631.15
Exceeding 50,000.00	35	17,500.00		

Source: Ghana Revenue Authority

Table 3: PIT Annual tax rate in Ghana-Effective 1st May, 2023:

Chargeable Income (GH¢)	Rate (%)	Tax Payable (GH¢)	Cumulative Income (GH¢)	Cumulative Tax (GH¢)
First 4,824.00	Free	NIL	4,824.00	NIL
Next 1,320.00	5	66.00	6,144.00	66.00
Next 1,560.00	10	156.00	7,704.00	222.00
Next 36,000.00	17.5	6,300.00	43,704.00	6,522.00
Next 196,740.00	25	49,185.00	240,444.00	55,707.00
Next 359,556.00	30	107,866.80	600,000.00	163,573.80
Exceeding 600,000.00	35	210,000.00		

Source: Ghana Revenue Authority

The taxation levels follow a progressive structure, spanning from 0 to 25 percent. Annual earnings of GH¢ 4,824.00 is freed from taxation. Presently, the lowest applicable income for taxation stands at GH¢ 6,144.00, taxed at a rate of 5 percent. Furthermore, an upper threshold tax rate of 35 percent applies to annual incomes exceeding GH¢ 600,000.00 (as outlined in Table 2). Individuals who are self-employed are obligated to remit income tax payments in four equal instalments, each corresponding to the conclusion of a quarter throughout the year – specifically, in March, June, September, and December. This arrangement encompasses sole proprietorships and is designed to ensure tax compliance. Corporate income tax pertains to

incorporated entities governed by the Companies Code (Act 179) of 1973 or other applicable legislation. It encompasses the taxes levied on the profits generated by these companies within a given fiscal year. There is a 25 percent tax rate in effect for this category. However, companies that have been listed on the Ghana Stock Exchange receive a reduced corporate income tax rate of 22 percent during the initial three years following their listing. Additionally, mining enterprises are bound to a commercial income tax rate of 35%, while companies operating within the hotel industry are subjected to a reduced tax rate of 20 percent, rather than 22% as of the year 2011.

Furthermore, a 20 percent reduction in the corporate income tax (CIT) rate is implemented for financial institutions that typically extend loans to agricultural enterprises. Similarly, rural and community banks, as well as free zone developers and enterprises, are subject to an 8 percent tax rate after a 10-year tax holiday. Additionally, both domestic and foreign companies conducting business within the country are assessed taxes on their taxable incomes, which encompass net profits, interest, royalties, and rental income derived from companies exclusively engaged in real estate development. The third tax category encompasses various elements such as royalties, dividends, rental income, pensions, and license fees. Rent tax applies to individuals earning rental income, calculated on the total amount earned during a fiscal year. This tax, which consists of an 8 percent rate applied to the gross rental income, is a last obligation. A five percent rate of mineral royalty is charged on those who mine natural resources from the surface or beneath the soil. Dividends are considered taxable liabilities and therefore subject to an 8 percent tax rate. However, dividends obtained from unit or mutual trusts are exempt from taxation.

Additionally, of equal significance, there are several other direct taxes under administration in Ghana, which include vehicle income tax (VIT), tax stamp, gift tax, capital gains tax, airport tax, and stamp duty. Gift tax pertains to a levy paid by the recipient on the cumulative value of taxable gifts received, surpassing a minimum of GH¢50.00 within the assessment year. At present, the gift tax rate varies from 5 percent to 15 percent. This tax applies to various assets, encompassing land, buildings, securities, shares, bonds, business assets, and monetary funds (including foreign currency). Nonetheless, certain gifts are exempt from taxation, such as those received by an individual through a will or intestacy, from relatives, from a religious institution for public benefit, and for charitable and educational purposes. Capital gains tax refers to the tax remitted on earnings resulting from the realization or sale of taxable assets. The applicable rate stands at 5 percent, applicable to gains surpassing GH¢50.00. This tax is levied on diverse assets, encompassing structures, land, shares of

resident companies, and business assets, including goodwill. However, shares issued on the Ghana Stock Exchange and capital gains derived from companies listed on the exchange are exempt from this tax.

Introduced in February 2005, the tax stamp is a quarterly levy levied on people working in the informal economy as independent contractors on a small-scale basis. To determine fair prices, a variety of company owners, including dressmakers, susu collectors, chop bar operators, and butchers, are grouped by kind and then further divided into classes or sizes. The Stamp Duty Act, 2005 (Act 689), which was later amended by Act 764 in 2008, provides the foundation for stamp duty. Stamp duty is applied on documents made with the intention of recording transactions, even though it is not a tax on transactions by itself. Consequently, it is considered a tax on documents or specific instruments with legal implications. Vehicle income tax (VIT) is a tax collected quarterly from commercial vehicle operators since its introduction in the country in July 2003. The tariff that applies differs according on the number of passengers and the kind of vehicle used for transportation, including tour buses, taxis and 'trotros'. VIT stickers fall into different categories and require quarterly purchases (Ghana Revenue Authority Act 791).

2.3.2 Indirect Taxes

These are types of taxes levied on the consumption of products and services within the country regardless of their origin. In Ghana, indirect taxes include VAT, customs and excise duty, petroleum tax, National Health Insurance Levy, and communication service tax (CST). A value-added tax is a broad tax levied on the costs incurred by customers when they make purchases of products and services. Value-added tax (VAT) is imposed on goods at every stage of their manufacture and distribution, and it eventually finds its way into the final product's price that customers pay. Businesses or authorised agencies apply the VAT tax at various stages, from manufacturing to retail, and collect it progressively. VAT, originally conceived by French economist Maurice Laure in 1954, initially targeted larger businesses but was subsequently expanded to encompass all business entities. Its adoption by the European Union, numerous African nations, Asia, and South America took place by the close of the century.

In 1993, the Ghanaian government acknowledged the necessity of VAT in its budget statement and commenced preparations for its implementation. The introduction of the VAT bill was discussed and endorsed by parliament in October 1994, with its enforcement postponed until March 1995. Following its successful implementation, the VAT rate stood at 12.5 percent

(Ministry of Finance, 1995). However, due to the enactment of the Value Added Tax Act, 2013 (Act 870), the VAT rate was increased to 15 percent, effective from January 2014. The VAT Act received presidential approval in December 2013 and was subsequently published in the Gazette during the same month. This amendment aimed to broaden the tax base by encompassing numerous profit-generating businesses that were operating outside the tax framework. Consequently, companies engaged in the production and/or supply of pharmaceutical products, beyond retail operations, were brought into the tax fold for the first time. Furthermore, gyms, spas, domestic airlines, and haulage companies were also included in this extended tax ambit.

The National Health Insurance Tax remains constant at 2.5%, for a total charge of 17.5 percent based on the taxable value of the delivered products or services. An excise tax of 20% is levied on the production or sale of manufactured goods, either at a fixed or ad valorem rate. Its purpose is to discourage the production and consumption of specific items. The administration of this tax falls under the domestic tax revenue division of GRA and is applicable to products such as beer, spirits, petroleum, and tobacco (Ghana Revenue Authority Act 791). Ghana's Education Trust Fund levy (GETFL) as of August 2018, is no longer part of the VAT, but it still applies at a rate of 2.5 percent on the VAT-exempt price of all VAT-able supplies. There is a comparable administration and reporting in revenue numbers to those of VAT and NHIL collects on imports for the levies collected. To support Ghana's Education Trust Fund, GETFL income are allocated. In 2021, the government introduced the COVID-19 Health Levy to support COVID-19 expenditures. It is effective from 1 May. This levy applies to the same taxable supplies as VAT and is not eligible for input tax deduction. Like NHIL and GETFL, this levy applies to the VAT-exclusive price of a good or service.

The Communication Service Tax is levied on payments for using the services provided by communication service providers. Consumers pay the communication service tax to these vendors, who then send the collected communication service tax to the government on a monthly basis. This tax applies to public and corporate data operators, national fixed network operators, broadcasting radio service providers, streaming television service providers, suppliers of internet services, mobile cell phone network operators, and free-on-air service providers. Petroleum tax pertains to the fee placed on oil companies. This taxation is implemented to account for economic rent stemming from land usage for mining purposes. (Ministry of Finance, 2020).

2.3.3 International trade taxes in Ghana

International trade fees are generated from charges on import and export operations. Import duty is a tax levied on all imports, with the exception of articles listed in the legislation as exemptions. This charge is levied on the Cost, Insurance, and Freight value of goods and varies in rate. Importantly, a clear differentiation exists between import VAT and import duty. Import VAT is imposed at a fixed rate and is calculated based on the value of goods inclusive of duty. Export duties are categorized into two main groups: traditional and non-traditional exports. Traditional export commodities encompass items like cocoa beans, logs, fresh fish, yam, electricity, and mineral ore (such as unprocessed gold). Conversely, non-traditional commodities include all goods not falling within the traditional category. Except for cocoa beans and hydrocarbons, the other exported commodities are subject to a zero percent duty rate.

2.4. Reforms in the tax system in Ghana

In 1983, the Ghanaian government initiated a series of fiscal and structural reforms with the goal of revitalizing economic growth. Among the significant changes implemented was the comprehensive restructuring of the tax system. The objective behind this undertaking was to amplify the revenue-generating capabilities of tax collection agencies, while simultaneously eliminating prevailing disparities and bolstering economic motivations. Moreover, multiple endeavours were made to augment both the effectiveness of administration and the fairness of the overarching tax framework. Consequently, the tax reforms progressed through three primary, interconnected phases: revitalization of the tax foundation, reinforcement of production incentives, and elevation of efficiency and equity within the tax administration. Bekoe, Danquah, and Senahey (2016) classified tax reforms in Ghana under three issues: broadening the tax base (1983-84), increasing production incentives (1985-86) and improving tax efficiency and equity (1986-2000).

Tax reforms in Ghana since 1984 have undergone several significant changes. Owusu-Gyimah (2015) provided an account of the evolution and reflection of tax administration and policy in Ghana:

- In June 1984, personal income tax rates were lowered, taxes on beer and cigarettes were raised.
- In 1985, the National Revenue Secretariat and two important revenue authorities, Customs, Excise, and Preventive Service and Internal Revenue Service formed as

independent entities outside the civil service. The National Revenue Secretariat managed Customs, Excise, and Preventive Service and Internal Revenue Service operations and advised the government on revenue policy. In addition, an investment code was developed, which provided tax breaks to investors, particularly in agriculture, industry, construction, and tourism.

- In 1987, the industrial sector's corporate tax rate was decreased from fifty-five percent to forty-five percent. Cigarettes, beer, alcoholic and non-alcoholic beverages all had special taxes applied to them. Excise duties were applied to all locally produced goods except petroleum products. A 10 percent sales tax on domestic electricity consumption was abolished, and import duty and purchase tax on all commercial vehicles were eliminated. Duty on basic primary resources and industrial goods was also reduced.
- In January 1988, the sales tax clearance certificate was reintroduced to ensure timely payment of sales tax and excise duties collected by manufacturers on behalf of the government.
- In January 1990, a super sales tax on luxury goods was introduced, ranging from 50 percent to 500 percent.
- In 1991, the super sales tax was reduced to a new range of 10 to 100 percent. Corporate taxes for agriculture, manufacturing, real estate, construction, and services were lowered to around 35 percent. The NRS was relocated under the Ministry of Finance, although it retained some degree of independence. The personal income tax threshold was raised from ₪126,000.00 to ₪150,000.00.
- In 1993, the contract for the design and implementation of the Value Added Tax (VAT) was signed. Petroleum taxes were increased to offset pay raises for civil servants and public service organizations in an election year. Prices of petrol, kerosene, and gas oil increased by approximately 60 percent, while LPG increased by about 20 percent.
- In 1994, a Debt Collection Unit was established under the Ministry of Finance in January. In December, the VAT bill was passed into law and became operational in March 1995.
- In 1995, the VAT became operational in March at a flat rate of 17.5 percent, replacing the previous 15 percent sales tax. However, in June, VAT was withdrawn, and the sales tax was reintroduced at a rate of 15 percent after mass demonstrations in almost all the regional capitals in the country.

- In 1996, the petroleum sector was deregulated, allowing competitive tendering between all oil trading enterprises and the Ghana National Petroleum Corporation (GNPC) for the procurement of crude oil and finished products to supplement the Tema Oil Refinery capacity.
- In 1998, the value added tax was reintroduced at a rate of ten percent, endorsed by Legislature in February. Taxpayer Identification Numbers were introduced to facilitate the assessment and collection of taxes from small businesses and market traders (Seth Terkper, 1995).
- In 1998, the VAT Service was established to oversee service and consumption taxes. Additionally, the Revenue Agencies Governing Board (RAGB) was established with the responsibility of supervising and coordinating the activities of revenue agencies and continuing the ongoing tax reforms.
- In 2002, a portion of revenue was retained to strengthen the semi-autonomous status of certain agencies.
- In 2004, a Large Taxpayer Unit was established on a pilot basis. It operated based on functional lines and introduced one-stop-shop operations to streamline tax-related processes.
- In 2006, the Ministry of Finance set up the Tax Policy Unit. It was in charge of gathering information, organising activities, and keeping a close watch on how tax laws were being applied.
- In 2009, there was an integration of all revenue agencies, a move that was included in the 2009 Budget as an official government policy. In December 2009, the Ghana Revenue Authority Act 791 was passed, providing a legal basis for the establishment of the Ghana Revenue Authority.
- In 2010, specific thematic areas for further tax reforms were identified, indicating the government's commitment to improving the tax system.
- In 2011, a Strategic Management approach was adopted, involving the development of Strategic and Modernisation Plans, which aimed to enhance the efficiency and effectiveness of tax administration and collection processes.

2.4.1 Broadening the Tax Base

From 1983 to 1984, changes to the exchange rate were implemented to enhance the revenue derived from cocoa export taxes along with import duties. Moreover, the increase in foreign exchange availability, stemming from donor inflows, facilitated an expansion in imports, thereby expanding the base for import taxes. As a result, the strategies for tax reform were predominantly devised to effectively revitalize the foundation of the tax system. This foundation had experienced a decline during this period due to the persistent overvaluation of the domestic currency and significant discrepancies between official and market prices (Kusi, 1998; Osei and Quartey, 2005). Furthermore, the overhaul of the tax system also sought to broaden the scope of taxation, curtail evasion, and mitigate the overall tax burden. Consequently, the implementation of a multiple exchange rate system in 1983 emerged from the imposition of surcharges on foreign exchange payments, alongside the provision of bonuses on foreign exchange receipts. By 1988, two distinct exchange rates were operational, which eventually underwent unification. Additionally, during the period of 1983-1984, the regulation of prices was deregulated. In order to maintain prices and provide producers with incentives for commodities like cocoa, rice, maize, and palm oil, this enabled consumer products pricing to take into account both production costs and profit margins.

Furthermore, budgetary subsidies for public utilities and consumer products were gradually eliminated. As part of the recently implemented cost recovery measures, changes were made to the public sector services domain, including higher taxes, levies, and charges. Furthermore, price changes and exchange rate adjustments triggered adjustments to the tax system's import taxes, sales, and purchase tax assessments. These tax regimes were reassessed in order to provide a foundation for items that are subject to duties; this included full face values in addition to designated surcharges. In the same way, actual income received in the most recent year was substituted for the previous year's profits for determining corporate income tax (Kusi, 1998). In addition, the practice of paying taxes in advance was discontinued at the start of each quarter, which had resulted in significant tax evasion. Rather, businesses and independent contractors were allowed to pay their taxes at the end of the quarter. In general, the minimum threshold for personal income exempt from taxes was increased, and this was done in tandem with a decrease in marginal rates to lower the average effective rates.

2.4.2 Enhancing Incentives for Production

Two major legislative initiatives were implemented as part of Ghana's second phase of tax reform: the Investment Code (PNDC legislation 116, 1985) and the mining Commission Law, 1986, a new legislation pertaining to minerals. This phase was to improve the tax structure even more. Four major investment industries were identified under the investment code: manufacturing, tourism, construction and real estate, and agriculture. Businesses operating in these industries were qualified for a variety of subsidies and tax breaks. For example, businesses with top priority that participated in authorised scientific research projects to further their business interests were eligible to a full tax credit for the associated capital costs incurred throughout the project. Additionally, to encourage the distribution of industrial activity throughout different regions, companies outside the Accra-Tema metropolitan area were eligible for income tax discounts ranging from 15 to 40 percent, contingent on the location.

Furthermore, the investment code incorporated special provisions aimed at reducing or deferring taxable income for enterprises, particularly in areas lacking basic infrastructure. In cases where an enterprise undertook the responsibility of providing such infrastructure, the code allowed for a reduction or postponement of the associated taxable income. The introduction of the new minerals law Minerals Commission Law 1986 resulted in the revision of eight pre-existing statutes, thereby enhancing the clarity of mining rights and introducing fresh incentives for potential investors. These incentives encompassed corporate tax allowances and capital allowances, enabling companies to offset anywhere from 40 percent to 100 percent of their capital investments against tax liabilities. Additionally, companies gained the authorization to utilize offshore bank accounts for activities like servicing foreign loans, disbursing dividends, and remunerating expatriate staff

2.4.3 Improving Tax Efficiency and Equity

After 1986, the scope of tax changes was broadened to include improving taxation performance and establishing fairness within the framework of taxation. Prior to 1986, there was insufficient control of the tax administration system, and assuring tax compliance was difficult. As a result, a critical component of the tax reform project was the strengthening of revenue collection agencies in order to not only increase revenue production but also restructure the tax system for enhanced efficacy and justice. Customs Excise and Preventive Service, Internal Revenue Service, and National Revenue Secretariat are three distinct independent bodies founded in 1985. Along with making recommendations to the government regarding revenue policy, the

NRS was entrusted with overseeing the operations of CEPS and IRS. To further enhance tax collection efficiency, the IRS and CEPS underwent a transformation in 1986, becoming standalone bodies with revamped administrative framework reminiscent of state-owned enterprises. Concurrently, fresh incentive policies were introduced to elevate productivity among the staff of IRS and CEPS. Moreover, during the period from 1986 to 1992, the IRS operated with ministerial powers in conjunction with the Ministry of Finance. However, this full autonomy was subsequently partially adjusted (Kusi, 1998).

In response to recommendations from the World Bank in 1989, the Ghanaian government embarked on the initiative to modernize the management information systems for tax administration through computerization. Additionally, they introduced a distinctive taxpayer identification numbering system. Regarding direct taxes, the corporate tax reform aimed to achieve a gradual reduction in tax rates, while also addressing distortions arising from the application of numerous tax regulations to diverse forms of capital financing and income from different sectors. In 1986, the corporate tax rate stood at 55 percent, except for mining and manufacturing, which faced a rate of 50 percent. The financial services industry witnessed a diminution in its tax rate from 50 percent in 1991 to 40 percent in 1992. By 1993, all corporate tax rates were lowered to 35 percent, excluding the mining sector. The reduction to 35 percent was a result of amendments to the Petroleum and Mining Law in 1994 (Kusi, 1998). Since 2006, the corporate tax rate has remained at 25 percent, coinciding with a decrease in National Reconstruction Levy rates. These reductions in the corporate tax burden aimed to position Ghana favourably among other attractive destinations for foreign direct investment (FDI) and to facilitate medium to long-term planning.

Significant adjustments were also made to the structure of the individual taxation of income scheme. In 1986, the existing 17 tax brackets were replaced with 5 brackets, each featuring new effective tax rates. These new rates brought substantial alleviation to households with lower incomes. The sixty percent top marginal tax bracket in the past, applicable to annual chargeable incomes surpassing 75,000 cedis, was diminished to 55 percent for taxpayers whose annual chargeable income exceeded 180,000 cedis (Kusi, 1998). The primary objective was to reduce the rates, aiming to minimize disincentives towards heightened productivity and entrepreneurial endeavours. The income tax, imposed under the Internal Revenue Act (Act 592), was approved in 2000 and became fully functional in 2001. In the inaugural year, the minimum taxable personal income was set at 1,200 cedis, subject to a 5 percent tax rate. The highest marginal tax rate, initially at 35 percent, was subsequently reduced to 30 percent for personal incomes surpassing 4,800 cedis. Furthermore, a subsequent decrease took the top

marginal tax bracket down to 25 percent for individual earnings exceeding 9,600 cedis in 2006. This structure remained in effect until November 2011 when a new tax rate configuration was introduced. The recent amendment by the Ghana Revenue Authority peaked the top marginal tax rate at 35 percent, and the threshold for personal income was raised to GHC600,000.00 in May 2023.

2.5. Challenges of the Tax Reforms

Tax reforms that lead to an increase in the tax burden borne by citizens are rarely well-received worldwide. Mahon (2000) argues that effective tax systems have historically emerged through the exchange of resources for institutions, with the intention of enabling governments to sustain tax efforts through the provision of capable institutions. While Ghana is not an exception to this general rule, the country has managed to minimize public resistance to tax reforms. This outcome can be attributed to the historical context of Ghana, where alternative viewpoints on policy implementation were often viewed as acts of sabotage due to the predominantly military rule until 1992. For instance, the transition from sales and service tax to VAT in the 1990s represented a substantial shift in the policy implementation landscape. Inadequate investment in education and institutional capacity contributed to the initial attempt to implement VAT in 1995 failed (Addison and Osei, 2001). The decision to switch from the 15% sales tax rate to the 17.5% VAT rate was made as a result of the absence of a thorough policy review. Furthermore, a lack of public education added to the misunderstanding surrounding the new system. Even with the advantages of transitioning from sales tax to VAT, there are still a lot of obstacles that limit the amount of money collected under the current tax rates. In Ghana, the institutional structures for tax collection still exhibit weaknesses, partially due to a lack of resources and skilled personnel capable of providing potential taxpayers with detailed information regarding the origins and levels of their income tax

2.6. Tax Revenue Composition in Ghana

According to the Ghana Revenue Authority report 2022, direct and indirect taxes, including import duties, constituted forty-eight percent and fifty-two percent of Ghana's tax revenues in 2020 and 2021 respectively. In 2019, Ghana achieved a tax-to-GDP ratio of 13%, generating GH45.6 billion in tax revenues. This figure remains lower than the typical tax revenue percentage in wealthier nations and is consistent with most Low- and Middle-Income Countries

(LMICs). For example, in 2018, the average tax revenue percentage among OECD nations was 33.9% (OECD, 2020). However, Ghana's tax collection, in comparison to its regional counterparts and countries with similar income levels, remains relatively lower. Given this context and in alignment with the government's objective to enhance domestic revenue mobilization through the Ghana CARES 'Obaatanpa' program, there is a medium-term goal to elevate the tax-to-GDP ratio to 20% by 2023 (Ministry of Finance, 2020).

2.7. Economic Growth

Economic growth is often measured by macroeconomic indicators. A key indicator to economic prosperity is a country's Gross Domestic Product (GDP). The GDP is a commonly used indicator of the value added generated in a nation during a specific time period by the production of goods and services. In an open economy, it includes net exports, government spending, private capital investment, and private consumption. This is the GDP equation:

$$Y = C + I + G + NX$$

In this case, Y stands for overall production, C for private consumption, I for all capital spending, G for government spending, and NX for net exports which can be positive or negative. Economic growth serves as the foundation for future prosperity and living standards across diverse countries. Even slight variations in growth rates can have a significant impact on future per capita GDP levels and living conditions, as GDP per capita is a very prominent measure of life standards in different nations. Economic growth is characterized by rapid shifts from exporting of raw materials production to value added goods. Growth in GDP or national income refers to an annual increase in the value of material produced (Atawodi and Ojeka, 2012). In the long term, numerous factors can limit economic growth, including population growth, resource constraints, insufficient infrastructure, inefficient resource utilization, excessive government intervention, challenging institutional and cultural norms, and more. Efficiency in resource allocation and expanding a country's production capacity are two strategies for achieving economic growth.

The ability to redistribute income of individuals and society, and even minor changes in growth rates can have a significant long-term impact. Income redistribution is more feasible in a dynamic, broadening society than in an inflexible one. Occasionally, economic expansion is disrupted by economic fluctuations (Abala, 2014). Implementing expansionary monetary and fiscal policies can help alleviate recessions and boost GDP beyond its potential level. Economic

growth implies a change in potential output due to alterations in the supply of factors like labour and capital, or improvements in the efficacy of these factors (output per unit of input).

The objective of economic growth is to elevate the overall financial well-being of the general populace, thereby resulting in an upswing in both national income and per capita income. In the realm of macroeconomics, the impact of alterations in tax policy on economic activity holds fundamental significance (Owusu-Gyimah, 2015). There is a widespread perception that taxes have a detrimental impact on economic growth, indicating an adverse link between taxes and economic expansion. This stems from the notion that higher taxes introduce distortions into the economy, consequently leading to reduced efficiency (Nonvide and Amegnaglo, 2017). Tax hikes serve as potent catalysts for individuals to recalibrate their lifestyles, as they realize that, irrespective of their approach, taxes will leave them worse off. The tax system of a nation wields an enormous influence over various other socioeconomic parameters.

To elucidate further, the interconnectedness between tax policy and economic growth and development prevails in both emerging and advanced economies alike (Ahmad et al., 2016). Different dimensions of taxation are intrinsically tied to economic development and growth. Taxation exerts a pronounced impact on economic decisions, thereby influencing the pace of growth. In simpler terms, heightened taxes result in diminished returns on investment, which in turn contributes to a slower growth rate for the economy. Taxation encourages the redirection of funds from the private sector to the public sector, and it can also be harnessed to create a conducive environment for private enterprises. Tax revenues play a crucial role in funding infrastructural development such as roads, enhancing security systems, and providing health and educational amenities across the nation. Effective management of taxation and public spending policies aimed at fostering efficiency and fairness within the economy hold paramount significance. Furthermore, astute economic policy seeks to optimize the benefits and drawbacks of the tax system to stimulate growth and enhance the overall social and economic well-being of the populace. Assessing the economic impacts of tax changes can prove intricate, particularly in terms of evaluating changes in real GDP or its components such as consumption and investment. Erroneous conclusions might arise from observing tax cuts during economic downturns, potentially leading to misconceptions about the impact of tax reductions on growth.

2.7.1 Economic Growth and Government Expenditure

The continued relevance of the relation between government spending and economic growth is a topic of controversy among investigators and lawmakers in the modern day (Ibrahim, 2019). Generally, there is consensus among researchers that public sector spending plays a pivotal role as a major instrument with significant influence over the economy's performance (Adu et al., 2013). Consequently, public expenditure serves as the primary means by which the collective needs of the citizenry are met by governmental authorities. Fundamentally, public expenditure encompasses the allocation of government funds derived from taxes and other revenue sources (Mallick et al., 2016). These expenditures primarily serve the dual purpose of sustaining government operations and facilitating the rapid growth of the economy (Antwi et al., 2013). Moreover, it functions as a fiscal tool designed to manage and allocate revenues effectively for the benefit of the nation. Government expenditures encompass a diverse range of areas within the economy, including investments in infrastructure, capital projects, pensions, and more (Owusu-Nantwi and Erickson, 2016). As asserted by Owusu-Nantwi and Erickson (2016), public expenditure represents both the inception and culmination of the government's revenue collection process.

Government expenditure tends to be separated into two main categories: current expenditure and capital expenditure. Current expenditure encompasses costs related to salaries, wages, labour, consumables, and similar items. In contrast, capital expenditure comprises investments in assets like hospitals, schools, roads, and other infrastructure projects that contribute to a nation's growth and development (Akanbi et al., 2019). There exists a direct correlation between the extent of public sector expenditure and a country's economic growth (Aziz and Asadullah, 2017). Consequently, substantial emphasis is placed on the role of public spending in addressing significant challenges, including unemployment, exchange rate volatility, inequality, inflation, fluctuations in oil prices, balance of payments stability, and most notably, economic growth (Aziz and Asadullah, 2017). In its ideal manifestation, a nation's economic growth is anticipated to enhance living standards by facilitating access to education, healthcare, infrastructure, housing, quality food, improved road networks, and more. These advancements are deemed essential and desirable for addressing human capital development issues and promoting key economic activities within a country.

However, in numerous sub-Saharan African nations, the comprehensive utilization of public expenditure as a fiscal policy tool and its effects on economic growth remain a subject of uncertainty. Onifade et al. (2020) underscored that during the preceding twenty years, public

spending has increased; nevertheless, doubts linger regarding whether these increments have translated into the anticipated economic growth in Nigeria. In Botswana, government expenditure has consistently been averaged by 40% of GDP between 2000 and 2016, with an upward trajectory (Oyinlola, 2019). There is still much to learn about how exactly government spending affects Botswana's growth. In a nation characterized by a limited government, an increase in the government's scale leads to higher returns and enhanced economic output (Afonso and Ibraimo, 2020). A nation's development can be strongly impacted by the influence of expenditures by governments on booting the economy. In this context, the production of public goods and services, which serve as inputs in private sector production processes, as well as improvements in both physical and human capital investment, can contribute positively to economic expansion (Chandio et al., 2016). However, when a government's size becomes excessively large, with financing relying on tax hikes, borrowing, and increased money printing to sustain its expenditures, it can adversely affect a country's economy (Afonso and Ibraimo, 2020).

Furthermore, the literature explicitly argues that an expansion in government size can create opportunities for lobbying, corruption, and other rent-seeking activities. These activities can lead to significant macroeconomic challenges, including a decline in the quality of public infrastructure services such as justice, healthcare, education, and defense, as well as reduced attractiveness for foreign investment and unfair income distribution (Wang et al., 2019). In Ghana, despite a consistent increase in government expenditure, there have been significant deviations in terms of economic growth over an extended period (Iheanacho, 2016). The government allocates substantial funds to sectors such as healthcare, education, public order, infrastructure, justice, national security, defense, and general administration. Theoretically, any rise in government expenditure in these sectors ought to spur economic expansion.

However, this is not the case in Ghana, possibly due to the government's allocation of funds to non-growth expenditures that do not stimulate economic growth. Consequently, the issue of government expenditures as a means to foster sustainable economic growth is of paramount importance and requires a thorough assessment (Nketiah-Amponsah, 2009). Ghana has experienced various episodes of economic instability in recent times, yet the underlying causes of these fluctuations remain inadequately understood. In particular, a thorough assessment of the interaction among expenditures of governments and economic growth has not been established. Studies have attempted to rigorously analyse the mechanisms through which different types of government expenditure can influence the economy.

2.8. Inflation and Economic growth

The connection between inflation and economic growth stands as one of the most pivotal macroeconomic debates among macroeconomists, policymakers, and central monetary authorities across nations. Diverse findings emerge from logical and empirical research into the association with inflation and economic growth. High inflation is recognized for its multitude of adverse effects: it places societal welfare costs, disrupts efficient resource allocation by obscuring the signaling function of relative price changes. Inflation further hinders financial development by increasing intermediation costs, disproportionately affects the impoverished as they lack financial assets that can hedge against inflation, diminishes a country's global competitiveness by elevating the relative cost of its exports, thereby adversely affecting the balance of payments, and, ultimately, slows the rate of over time development of the economy (Ocran, 2007).

Conversely, the literature widely agrees that moderate inflation can actually promote economic growth (Adu, 2013). Inflation may stimulate short-term growth through expansionary macroeconomic policies, considering this, such an effect cannot persist in the course of time. As indicated by (Barro 1995), the link between inflation and growth is categorically negative over the long term. Inflation exerts a detrimental impact on growth and progress by introducing economic distortions and uncertainty, impeding the ability to engage in long-term planning, and diminishing savings and capital accumulation. In times of inflation, a set sum of money can procure a reduced quantity of goods and services compared to what it would have purchased otherwise. This leads to a significant erosion of the real value of money, thereby diminishing the purchasing power of money for consumers (Gyebi et al., 2013).

2.9. Theoretical Review

Certain scholars have put forth theoretical frameworks that elucidate the connection between economic growth and its drivers. These models are separated under two overarching categories: Exogenous growth theories and Endogenous growth theories.

2.9.1 Exogenous Growth Theories

Theories of exogenous growth include the initial notions that suggest growth is a function of capital accumulation or enhancements in the productivity of capital stock due to externally assumed technical progress (Kurz and Salvadori, 2014). Notable examples of classical

exogenous growth theories in the literature include those formulated by Gustav Cassel (Cassel, 1932) and Knut Wicksell (Wicksell, 1934). Within the realm of neo-classical exogenous growth theories by Ricardian theory (Ricardo, 1951) and the Solow-Swan growth model (Solow, 1956; Swan, 1956). These exogenous growth theories, due to their primary focus on capital accumulation as a growth driver, assume that the assumption of falling marginal rates of capital is the primary reason why growth is constrained.

Determining technology, which comes from the aggregate production function, is the fundamental component in the Solow growth model cycle. This model establishes the relationship between output per worker (y) and capital per worker (k) alongside the influence of technology (A) through the equation:

$$y = Af(k),$$

where output per worker is dependent on capital per worker, as indicated by $f(k)$. Similar to the production functions, technology augments output and it is assumed that the function exhibits characteristics where increased capital yields greater output per capita at a diminishing rate. The subsequent element involves the accumulation of capital. In the scenario that k_t represents the capital per person at the beginning of year t , the relationship is as described below:

$k_{t+1} = k_t (1 - \delta)$. The aforementioned equation illustrates how the capital stock varies with time. δ represents the physical depreciation rate, meaning that δk_t units of capital lose their value due to depreciation from year t to year $t+1$; however, during the year t , there is an injection of investment (it) that leads to the creation of new capital for the next year. Savings is the Solow growth model's last component. Savings and investing are intertwined in the accumulation equation because they are identical in a closed market. Assume the following description of saving per capita (s_t):

$$s_t = s \times y_t.$$

This time around, the constant's', which stands for the percentage of the overall output that has been saved, ranges from zero and one ($0 < s < 1$). Investment and the capital level can be linked by establishing a connection between savings and investment and the per capita production function.

It is equal to $sAf(k_t)$.

As a result, the formula controlling the growth of the capital stock has the following structure:
 $k_{t+1} = k_t (1 - \delta) + sAf(k_t)$.

Once the function $f(k)$ has been delineated, a trajectory of the capital stock across time can be traced. Typically, the temporal course of the capital stock possesses two noteworthy characteristics:

Steady state: There exists a specific magnitude of the capital stock such that if the economy amasses that quantum of capital, it stabilizes at that particular level—referred to as the steady state capital level, symbolized by k^* .

Stability: The economy is prone to focusing on the per capita capital stock k^* . To be more precise, the economy accumulates capital to reach k^* when the current capital stock $k_{t+1} > k_t$. Conversely, if the present capital stock surpasses k^* , the economy dissipates capital to attain $k_{t+1} < k_t$.

The following equation represents the steady state capital level precisely:

$$k^* = k^*(1 - \delta) + sAf(k^*).$$

The capital lost to depreciation precisely balances savings in steady state. Net investment consequently approaches zero. The stability property indicates that the economy will increase capital accumulation so that $k_{t+1} < k_t$ in the event that the current capital stock drops below k^* . According to this, the economy will exhaust its capital holdings if the existing capital stock reaches k^* , which will result in $k_{t+1} < k_t$.

Tax policy has virtually little effect on long-term economic growth, according to Solow's (1956) neoclassical model. The main components of production in this model—labour and technical advancement—are presumed to be determined externally to the model's framework. If different tax types have different effects on the equilibrium capital-labour ratio in the setting of the neoclassical Solow growth model, the per capita earning level at constant level could fluctuate depending on the tax policy measures adopted. Tax policy considerations also come into play for short-term growth as the economy converges to its equilibrium state. While economic growth fundamentally hinges on externally driven technological advancements, the transitional process can extend over multiple decades once the economy attains its steady state. Thus, there is little room for the influence of tax policy decisions under the Solow model.

2.9.2 Theories of Endogenous of Growth

Endogenous growth models provide a framework for attaining enduring growth over time. In contrast to exogenous growth models, where sustained growth is limited by diminishing marginal returns to capital, endogenous growth models offer the potential for continuous expansion. Additionally, exogenous growth models presuppose that technical progress—such as technological advancements and increases in human capital or education—occurs externally. Researchers' attempts to uncover the sources of technical progress gave rise to the emergence of endogenous growth models. Romer (1986) developed the AK model, which is the initial illustration of an endogenous growth model. Within the AK model, production relies solely on a single input, with constant returns to that input (capital) being maintained. This signifies that continuous growth can be attained through a consistent augmentation of the capital stock (Myles, 2009b). In essence, tax policies that foster increased savings, subsequently driving greater investment in the capital stock, might potentially yield a positive impact on economic growth.

Romer (1990) and Mankiw et al. (1992) made modifications to the AK model by incorporating human capital as a second input into the original production function. This new version was termed the "human capital approach" to achieving endogenous growth. Human capital is tied to labour quality or skill. This model assumes that human capital, akin to physical capital, can be cultivated and accumulated through education and training over time. By introducing human capital, the level of technical progress is determined by the quality of labour. Within the human capital model, tax policies that encourage education are anticipated to bolster economic growth. Moreover, Schumpeter's (1934) notion of creative destruction served as the inspiration for endogenous growth models that are focused on innovation. Creative destruction involves the introduction of superior goods, services, or processes that replace outdated ones. Segerstrom et al. (1990) are recognized for pioneering the modelling of the growth-innovation interplay. In this innovation-focused model, technical progress is embodied in the advancement of superior inputs that replace outdated ones with improved products and processes. Essentially, productivity escalates as superior inputs supplant older ones. The implication for tax policy is that taxes fostering innovation have the potential to enhance economic growth.

2.9.3 Empirical Review

The influence of tax policy on economic growth exhibits variation across countries, exhibiting both immediate and long-term consequences. Numerous empirical investigations have employed growth models with diverse specifications to scrutinize the theoretical postulations and the extent of impact of tax policies across different countries and regions. The connection between taxation policies and growth was initially established in the mid-1980s through the application of the neoclassical growth model, often associated with a single good and an indefinitely lived population (Lucas, 1985 and Skinner, 1988). The results prove that taxes have no lasting effect on economic progress. This is because external forces like rising populations and advances in technology dictate the steady-state expansion of output. Nevertheless, growth rates can be influenced during the transitional phase between these two stable equilibrium states.

Harberger (1964) delves into the relationship between taxation policy and growth, suggesting that the arrangement of direct and indirect taxes significantly influences investment and growth theoretically. Nonetheless, in practice, the impact of taxation policy on growth appears to be minimal. Harberger further evaluates that alterations in taxes are unlikely to elevate the growth rate of national income by more than 0.1 to 0.2 percent (Mendoza et al., 1995). From Harberger's standpoint, changes in taxation policy seem to hold little sway over the economy's growth, essentially rendering taxation policy "super neutral." Mendoza et al. (1995) employed endogenous growth models to examine Harberger's findings. Their research aligns with Harberger's assertion that taxes exert a marginal impact on growth. In practical terms, significant adjustments to the tax system are necessary to manifest observable shifts in economic growth. However, they do not account for the fact that Harberger's concept of super neutrality does not imply that tax reforms are ineffective. In truth, reducing tax distortions can substantially enhance overall welfare (Mendoza and Tesar, 1995).

Engen and Skinner (1992) delineate five discrete mechanisms through which taxes could promote the economy's expansion. Initially, substantial taxes could reduce investment and prevent the capital stock from overall advancement. This may show up as higher statutory tax rates on individual and corporate income, higher effective income tax rates, and lower maintenance credits. Consequently, taxes possess the ability to curtail the expansion of the labour pool. Reductions in working hours or measures that discourage people from joining the workforce are both instances of this. Taxes could impede the pursuit of education, training, and skill development, potentially hindering the growth of the labour force by influencing career

choices. Moreover, tax policies might discourage improvements in productivity. The positive ripple effects on productivity stemming from these endeavours could be dampened by limiting investment in research and development (R&D) and constraining venture capital support for technology-driven enterprises. By enhancing the productivity of existing labour and capital, these ripple effects may ultimately foster economic growth.

Esterly (1993) investigates the influence of taxes on growth from a unique perspective. Rather than directly examining tax rates, Esterly focuses on the distortions engendered by these tax rates. The study conducted by Summers and Heston in 1988, utilizing 1980 price data for 151 commodities across 57 nations in comparison to the US, serves as the basis for deducing these distortions. The extent of price fluctuations within each nation is subsequently utilized as an indicator of the proportional level of distortion resulting from taxes, price-controls, caps, and various interventions in those economies. The findings presented indicate that, even when accounting for other growth factors like initial country income and school enrolment, the variability in input prices significantly influences development. There are two enormously shortcomings to this method, though. First, it is believed though not proven beyond a reasonable doubt that the price variance serves as a reliable indicator of the level of economic distortion. Second, it is not always easy to translate the effects of price changes into the implications of changing tax rates. Understanding how taxes affect market equilibrium, which in turn affects pricing, is necessary for this translation.

Furthermore, in their study, Milesi-Ferretti and Roubini (1995) utilize an endogenous growth model to explore the impacts of the existing taxation system on economic growth in the USA. Their analysis primarily centres on the taxation of income and consumption. According to their research, taxing factor income resulting from both physical and human capital drags down growth. This is due to the fact that the introduction of taxes diminishes the rate of return from factor inputs, thereby discouraging the accumulation of labour and capital. In addition, the researchers propose that the impact of consumption taxes on growth is not negligible; Instead, it hinges on the flexible nature of the workforce supply. In essence, the more elastic the labour supply, the more likely consumption taxes will induce workers to substitute their work and education for leisure. Economic growth is consequently impeded as the labour factor's accumulation declines. However, their model suggests that this is the sole distorting impact that consumption taxes wield on growth. On the other hand, income taxes not only influence the trade-off between work and leisure but also hinder capital accumulation and economic growth through alternative mechanisms. These insights underscore that an optimal tax structure should be predominantly built upon consumption taxes rather than income taxes.

Gemmell et al. (1999) utilised panel data covering 22 OECD nations from 1970 to 1995 to undertake an analysis to gauge the impact of different tax types on per capita income growth rates. Their estimations, employing both pooled OLS and fixed-effects methodologies, reveal that higher tax levels correlate with a reduction in per capita income. They stress that the potential biases in the estimated effects of various tax types could emerge if other budgetary components like expenditures are omitted from the model. Suggested by Arnold et al. (2011), a pragmatic approach involves scrutinizing the growth consequences of revenue-neutral shifts in tax policy. This method avoids the difficulty of having fluctuations in public spending intertwined with changes in overall tax collection. To elaborate, Arnold et al. (2011) examines the long-term impacts of revenue-neutral tax policy changes on economic development using panel data from the OECD Tax Revenue Statistics, which cover 51 OECD nations between 1970 and 2004.

According to Arnold et al.'s (2011) Pooled Mean Group estimates, raising the overall tax revenue in the direction of consumption and real estate taxes while improving the composition of individual taxes is associated with higher long-term per capita income. Property tax is the most competitive option when it comes to a "tax and growth ranking," which considers the contribution of different tax forms on the amount of per capita income over time. Next in order are corporation income taxes, personal income taxes, and consumption taxes. The study highlights that personal and corporate taxes, along with immovable property taxes, have the least detrimental impact on GDP per capita over the long run. Additionally, Arnold et al. (2011) reveals that a strategic reduction in income tax for low-income earners can bolster economic growth and facilitate economic recovery. Such a reduction stimulates demand, mitigates income inequality, and amplifies work incentives. Besides, their results hold up well to robustness tests with different model specifications that include other independent variables such as the rate of inflation.

Employing annual survey data from 22 economies which are part of OECD between 1970 and 1995, Kneller et al. (1999) assessed the effect of tax regulatory initiatives on socioeconomic advancement. Their method involved defining the government budget limit in detail, including revenue and expenditure. This is in contrast to other endogenous growth models that just take the revenue component into account. The authors extended the methodology of Barro (1990), in which biases inherent in a partial specification are corrected by the comprehensive definition of the government's budget restriction. In their approach, rates of economic growth are regressed for both fiscal and nonfiscal variables. The analysis's conclusions show that increasing direct taxes significantly slows growth in contrast to

consumption taxes, which have a less severe negative impact. It is crucial to note that average tax rates, as opposed to marginal tax rates, are inevitably used in the estimation.

To examine the effects of tax laws on macroeconomic activity in Barbados, Greenidge and Drakes (2009) used an unrestricted error correcting model. This method is based on the work that was first presented by Pesaran and Shin (1997). It includes a general autoregressive distributed lag model that can account for both short and long-term impacts. This methodology addresses issues with small sample sizes and dynamic biases and has the ability to analyse co-integrated variables independently of their sequence of integration. The researchers use conventional growth variables to estimate this model in the study before introducing individual tax indicators. Primary constituent analysis and the calculation of tax indices are used to get these tax indicators. The results imply that direct taxes have a detrimental effect on both immediate and future growth. All taxes, however, only appear to have a short-term contractionary effect on the economy; they have no discernible long-term effects.

The impact of tax policy decisions on the post-World War II expansion of the US economy is examined by Romer and Romer (2010). They use a regression technique in which tax laws derived from narrative sources, such as presidential addresses, are regressed against changes in real GDP between 1947 and 2007. The modifications made to the tax code can be divided into two groups: those motivated by outside forces and those connected to future economic conditions. The outcomes of their study unveil a substantial impact of tax policy actions on output, resulting in a reduction from 3.0% to 2.5%. Moreover, it's observed that the effects on output are more closely linked to actual changes in taxes when compared to the implications of news about future alterations in the tax structure.

Cashin (1994) explores the impact of taxes, public investment, and public transfers on the economic growth rate through an endogenous model. The model reveals that distorting taxes exert a substantial adverse influence on growth. This phenomenon stems from taxes diminishing the marginal return on private capital, consequently hampering economic growth. In contrast, productive public expenditure, manifested in the form of public investments and transfer payments, serves to invigorate growth. Cashin's analysis concludes that in countries characterized by a smaller-scale state, the dominant effect of public investments on economic growth is positive. On the other hand, for nations with a larger-scale state, the prevailing impact is the reduction resulting from the adverse influence of distorting taxes on growth. Widmalm (2001) looks into the increasing repercussions of income-neutral modifications to tax policy using data from the OECD Tax Revenue Statistics covering 23 OECD countries from 1965 to

1990. The author separates taxes into five categories: corporate income taxes, individual labour and capital income taxes, property taxes, taxes on sales and utilization, payroll and retirement fund taxes.

Using the Analysis of Extreme Boundaries According to Leamer (1983), there is a strong inverse link between the proportionate of individual income tax and the rate of increase in per capita Gross Domestic Product. He did, however, disclose that there is a positive, albeit shaky, relationship between corporate income taxes and growth as a percentage of total tax collection. In a similar vein, the evidence on taxes on property, products and services, payrolls and social security contributions, and both is shaky. The estimation discovery, which considers investment to GDP ratio, indicates that the tax type may have an impact on growth through mechanisms other than the accumulation of physical capital, like availability of labour or the improvement of human capital. Using quarterly data from Jamaica, Scarlett (2011) uses an autoregressive distributed-lag model to assess the impact of tax policy on economic development.

Nguyen et al. (2021) conducted an examination of the impacts of individual income taxes, corporate taxes, and consumption taxes in the UK spanning the years 1973 to 2009. Their study amalgamates individual and corporate incomes, revealing a noteworthy influence on GDP, private consumption, and investment. Notably, income taxes are shown to enhance GDP by 0.78 percent when the average tax rate experiences a one percent reduction. While the results from consumption tax reductions did not yield statistically significant findings, the report does underscore the influence of transitioning from an income-based to a consumption-based tax structure on economic growth. It's observed that consumption taxes, in comparison to other tax forms, are generally considered less distortionary due to their limited impact on labour and investment incentives.

Stoilova (2017) undertook an investigation into the relationship between tax structure and economic growth across the 28 EU countries during the period 1996 to 2013. Employing Barro's endogenous model, the researcher identified that total revenue indeed exerts an impact on the economy. Interestingly, the study highlights that, in contrast to corporate taxes, personal income taxes have a positive influence on economic growth. In another study, Badri and Allahyari (2013) utilized the autoregressive distributed lag model (ARDL) to analyse the interplay between taxation, GDP, and employment in Iran from 1976 to 2007. Their findings suggest that taxes yield a substantial and negative impact on employment over the long term, whereas GDP has a positive and significant influence in the short term. Additionally, Zellner and Ngoie (2015) performed a Marshallian macroeconomic model analysis focusing on the

United States during the years 1987 to 2008, investigating the effects of taxes on economic growth. The outcomes of their study reveal that corporate taxes are associated with a detrimental impact on economic growth.

Gachanja (2012) conducted an examination of the interplay between Kenya's economic development and its taxation policies throughout the span of 1971 to 2010. The study's findings reveal a harmonious linkage between socioeconomic expansion and taxes. Specifically, there exists a positive correlation between the gross domestic product (GDP) and the diverse tax categories, with income tax demonstrating the most significant influence. Through Gachanja's (2012) application of the Granger Causality test, noteworthy patterns emerge. A reciprocal relationship between economic growth and excise tax is observed, while a unidirectional connection links income taxation and economic growth, as well as economic growth and VAT. Different routes of causality in these interactions are suggested by the analysis.

Moreover, Gachanja (2012) emphasizes that the varying utilization of tax funds may exert diverse impacts on economic growth. It is important to acknowledge that the model does not incorporate other variables influencing GDP beyond taxes, such as government spending and investment. In the context of this thesis, recent studies directly relevant to the subject shed light on the growth implications of tax policy. Ojede and Yamarik (2012) employed Error Correction Models to investigate the intricate long- and short-term relationship between tax policies and economic development in the United States. Their study encompassed data from 48 states spanning the years 1967 to 2004. The results showed that individual taxation of income had a beneficial impact on socioeconomic expansion, but business taxes have a negative one. Dackehag and Hansson (2012) used panel data from 1975 to 2010 in a parallel study to assess the interaction between income taxation and economic growth across 25 affluent OECD countries. Their findings indicate that both company and individual income taxes hinder economic growth. Nevertheless, they highlight a more robust connection between corporate profits and the expansion of the economy.

3. METHODOLOGY

This chapter outlines the development of the econometric model in relation to this thesis. It offers a comprehensive view of the methodology employed, data sources utilized, estimation technique and the specifications of the model.

3.1 Overview of Methodology

Tax revenue plays a significant role in influencing economic growth, a topic extensively explored by economists and researchers using various methodologies such as classical linear regression and panel data analysis. The analysis of time series technique is used in this study to gauge the optimal interaction of the parameters chosen. Although there are many models that look into long-term stable equilibrium relationships between variables, it's important to recognise that exogenous shocks might cause short-term changes (Shao et al., 2019). In regard to this, the Vector Error Correction Model was selected for the study in order to accommodate for the short-term dynamics and adaption process that occurs when parameters differ from their long-lasting state of stability. This model has an error correction term which is invaluable for modelling the immediate impact of shocks and understanding how the system reverts to equilibrium. In addition to measuring the long-term relationship between taxes and economic growth, the Vector Error Correction Model is designed to handle time series data that is non-stationary, which is typical in the field of economics. As a powerful instrument for modelling and understanding the dynamics between variables, the Vector Error Correction Model is a good choice for researching economic linkages, creating forecasts, and analysing policy. This thesis utilises a variety of techniques, including the Granger causality test, the Johansen test, the Augmented Dickey-Fuller unit root test for stationarity evaluation, the Vector Error Correction Model for cointegration analysis, and VAR lag analysis. To figure out the most suitable lag lengths for model estimations, the VAR lag analysis is necessary. Key insights into the conduct of the time series dataset can be obtained by analysing stationarity. The cointegration test assesses the overtime dependability of two variables, and the Error Correction Model validates that dependability. A description of the methodology is given in Figure 1, which is the pictorial overview at a glance.

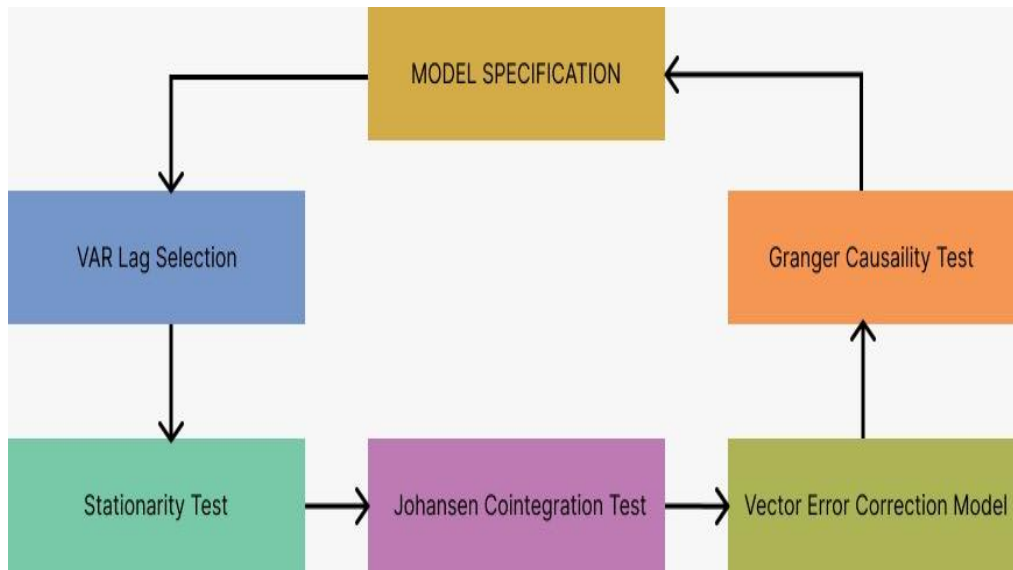


Figure 1: Chart flow of Methodology

Source: Authors own design

3.2 Data Collection and Variables

This study examined how tax policy is linked to economic growth using time-series data. The dataset consisted annual data spanning from 1990 to 2022 taken from three official sources namely: Ghana Revenue Authority (GRA), Ministry of Finance and Economic Planning, and World Development Indicators (WDI). The data set involve Gross Domestic Product (GDP) growth rate, total tax revenue consisting of direct and indirect taxes in billion cedis, capital expenditure and revenue expenditure in billion cedis as well as inflation rate. The gathered data is structured into dependent and independent variables for the purpose of evaluating the influence of economic growth on tax policy. In this research, the GDP growth rate has been utilised as a proxy to gauge the dependent variable. At market prices based on constant local currency, the GDP growth rate is the yearly percentage increase in GDP. The independent variables include tax revenue both direct and indirect, capital expenditure, revenue expenditure and inflationary rate. Secondary data sources were selected because they had undergone prior scrutiny by experts and regulatory bodies before their publication. While the trustworthiness of the secondary data used was unquestionable, the possibility of random errors cannot be entirely ruled out. Summary of the variables, expected signs of their coefficient and data source is depicted in Table 4.

Table 4: Overview of the variables, probable signs of their coefficient and data origin

Variable	Measurement/Proxy	Expected Sign	Data Source
Economic Growth	Real GDP growth (annual %)	Positive (+)	World Development Indicator
Inflation	Consumer prices (annual %)	Negative (-)	World Development Indicator
Total Tax Revenue	Direct and Indirect taxes (GH¢ billion)	Positive (+)	Ghana Revenue Authority (GRA)
Capital Expenditure	Consumer prices (GH¢ billion)	Positive (+)	Ministry of Finance (MOF)
Revenue Expenditure	Consumer prices (GH¢ billion)	Negative (-)	Ministry of Finance (MOF)

Source: Author's computation

3.3 Estimation Technique

The study employed a three-step approach in econometric analysis to validate the suitability of the VECM methodology. These stages involved employing various tools: a unit root test, the Johansen cointegration test, and assessing both immediate and long run impacts using the VECM approach. Establishing the variables' stationarity was the main goal of the initial phase. This step was essential to gauge both the long-term and short-term relationships within the model. Testing for variable stationarity is vital since the characteristics of these variables can significantly influence the model's outcomes. A time series is deemed stationary when its variance and mean exhibit stable statistical features, meaning neither of them have a unit root. The term "integrated of order 1" $I(1)$ is applied if, on the other hand, it becomes stationary following first-order differencing.

The preferred approach for assessing the stationary patterns of time series variables is the Augmented Dickey-Fuller test. The Johansen co-integration test was applied to confirm the co-integration of the variables under consideration as the investigation proceeded on to the second phase. Co-integration suggests that these variables could have a persistent, linear connection. According to Engel and Granger (1987) and Johansen (1988), this relationship

results in disequilibrium errors that tend to float about a mean of zero. The final phase of the investigation employed the Granger causality test. Determining which way tax policy and economic growth are causally related was the goal. This test establishes a causal relationship by reviewing if modifications in one variable leads to alterations in another. In summary, this study employed a comprehensive approach involving unit root tests, Johansen co-integration tests, and Granger causality tests to ensure the validity of the VECM methodology. These stages of analysis were crucial in establishing the relationships, trends, and influences among the variables under investigation.

3.4 Model Specification

As posited by the thesis's objective, there exists a correlation between tax policy and the economic growth of Ghana. In order to address the research inquiries at hand, this thesis employs a comprehensive macroeconomic model to evaluate the connections between specific variables. Within this context, the Gross Domestic Product (GDP) assumes the role of the contingent variable. Simultaneously, inflation rate, capital expenditure, revenue expenditure and tax revenue serve as independent variables, all represented within equation 1.

$$GDP_t = f(\text{infla}_t, \text{Capex}_t, \text{Rex}_t, \text{Ttr}_t) \quad (1)$$

In the equation, **GDP** signifies Ghana's Gross Domestic Product, **Infla** represents the inflation rate, **Capex** denotes the government's capital expenditure, **Rex** stands for the government's revenue expenditure and **Ttr** indicates the total tax revenue of the economy. The variable '**t**' denotes time. One technique used to forecast future events is time series analysis, which is based on past data and conditions. The time series econometrics model is frequently seen as a structural model in this context because it draws support from well-established economic theories (Loves et al., 2021). In the predictive formulations of many economic models, the descriptive variables include lagged values. For example, infrastructural investment necessitate time for realization. Consequently, the impact of such public investment on the growth of gross national product takes time to materialize and is anticipated to extend over several years (Baltagi, 2008).

The drawback of equation (1) is that it cannot assess the variables' long and short-term effects at the same time. In order to solve this, a Vector Error Correction Model (VECM) must be created from equation (1) which enables the comprehensive assessment of both the immediate and prolonged relationships. VECM primarily assists to evaluate how much a causal

connection emerges amongst the variables. However, it's important to note that once co-integration is established among the variables, VECM can be applied in such circumstances. As per Lupiski's findings (2013), when the variables demonstrate co-integration at order one I (1), the relationship between them should be presented using the VECM format, as illustrated in equation (2). The benefits of employing VECM are multifaceted. Firstly, the incorporation of an error correction term alleviates the challenges linked to multicollinearity, often prevalent in time series analyses. Secondly, the long-term impacts are effectively summarized within the level matrix. Lastly, interpreting VECM outcomes is notably more straightforward, as highlighted by Juselin (2006).

$$\Delta Y_t = \beta_0 + \sum_{i=1}^k \beta_1 \Delta Y_{t-i} + \sum_{i=1}^k \delta_i \Delta X_{t-i} + \varphi_i Q_{t-1} + \varepsilon_t \quad (2)$$

The long-run cointegration regression's residuals of the ordinary least squares are the source of the error correction term, also known as Q. Conversely, when GDP is taken into account as the dependent variable in equation 1, the Vector Error Correction Model can be written as:

$$\Delta \ln GDP_t = \alpha_1 + \sum_{i=1}^k \beta_{1i} \Delta \ln GDP_{t-i} + \sum_{i=1}^k \beta_{2i} \Delta \ln Infla_{t-i} + \sum_{i=1}^k \beta_{3i} \Delta \ln Capex_{t-i} + \sum_{i=1}^k \beta_{4i} \Delta \ln Rex_{t-i} + \sum_{i=1}^k \beta_{5i} \Delta \ln Ttr_{t-i} + \varphi_1 ECT_{t-1} + \varepsilon_{1t} \quad (3)$$

Despite the availability of a fundamental equation, the vector error correction model provides an elaborate testing technique. This mechanism adopts the autonomous variables to be the independent variables in various contexts. This reasoning allows for the deduction of four equations from the original single equation as follows:

$$\Delta \ln Infla_t = \alpha_2 + \sum_{i=1}^k \beta_{11i} \Delta \ln Infla_{t-i} + \sum_{i=1}^k \beta_{12i} \Delta \ln GDP_{t-i} + \sum_{i=1}^k \beta_{13i} \Delta \ln Capex_{t-i} + \sum_{i=1}^k \beta_{14i} \Delta \ln Rex_{t-i} + \sum_{i=1}^k \beta_{15i} \Delta \ln Ttr_{t-i} + \varphi_2 ECT_{t-1} + \varepsilon_{2t} \quad (4)$$

$$\Delta \ln Capex_t = \alpha_3 + \sum_{i=1}^k \beta_{21i} \Delta \ln Capex_{t-i} + \sum_{i=1}^k \beta_{22i} \Delta \ln GDP_{t-i} + \sum_{i=1}^k \beta_{23i} \Delta \ln Infla_{t-i} + \sum_{i=1}^k \beta_{24i} \Delta \ln Rex_{t-i} + \sum_{i=1}^k \beta_{25i} \Delta \ln Ttr_{t-i} + \varphi_3 ECT_{t-1} + \varepsilon_{3t} \quad (5)$$

$$\Delta \ln Rex_t = \alpha_4 + \sum_{i=1}^k \beta_{31i} \Delta \ln Rex_{t-i} + \sum_{i=1}^k \beta_{32i} \Delta \ln GDP_{t-i} + \sum_{i=1}^k \beta_{33i} \Delta \ln Capex_{t-i} + \sum_{i=1}^k \beta_{34i} \Delta \ln Infla_{t-i} + \sum_{i=1}^k \beta_{35i} \Delta \ln Ttr_{t-i} + \varphi_4 ECT_{t-1} + \varepsilon_{4t} \quad (6)$$

$$\Delta \ln Ttr_t = \alpha_5 + \sum_{i=1}^k \beta_{41i} \Delta \ln Ttr_{t-i} + \sum_{i=1}^k \beta_{42i} \Delta \ln GDP_{t-i} + \sum_{i=1}^k \beta_{43i} \Delta \ln Capex_{t-i} + \sum_{i=1}^k \beta_{44i} \Delta \ln Rex_{t-i} + \sum_{i=1}^k \beta_{45i} \Delta \ln Infla_{t-i} + \varphi_5 ECT_{t-1} + \varepsilon_{5t} \quad (7)$$

The goal is to evaluate departures from every possible direction so that this thesis can be interpreted objectively. In order to avoid interpretational bias, the VECM accepts all possible outcomes. Here, Δ stands for the differencing operator, α for the constant that goes with each equation, and $ECT_{(t-1)}$ for the one-period lagged error term. The coefficients that need to be estimated are indicated by the symbol β , and the variable k represents the lag length. As the disturbance terms, the ε . variables show a serial lack of correlation. Error correction term must exhibit statistical significance and negative in order for the dynamics to settle to the equilibrium state over a long time (Adefeso et al., 2013). The model's present outcomes are influenced by past equilibrium when the error correction term is significant and adverse.

Using the term error-correction to characterise the impact of the preceding period's diversion from the long run equilibrium on the short-run variable is proper. Error correction term coefficient φ is the pace of adjusting concept. It measures the speed at which alterations in the reliant variable cause the target variable to return to equilibrium. The Augmented Dickey-Fuller test, originally introduced by Dickey and Fuller (1979) and Dominic et al. (1992), was employed in the study to evaluate the stationarity attribute of the time series data. Granger (1981) noted that there are a variety of approaches to time series analysis that can be used to analyse economic data. Furthermore, the research utilised Johansen's (1988) methodology to determine whether co-integration or a long-term link existed between the variables under consideration. According to Adamec (2014), cointegration is an essential component in reducing the possibility of finding a misleading regression link between these variables. Additionally, a two-equation test known as the Granger causality test was utilised in relation to the variables selected for this investigation. In the initial regression equation, a continuous causation that is unidirectional involving two specific time series variables is investigated. The mathematical formulae for these equations can be stated as follows:

$$x_t = C_1 + \sum_{i=1}^k a_1 y_{t-i} + \sum_{i=1}^k \beta_1 x_{t-i} + \varepsilon_t \quad (8)$$

$$y_t = C_2 + \sum_{i=1}^k a_2 y_{t-i} + \sum_{i=1}^k \beta_2 x_{t-i} + \varepsilon_t \quad (9)$$

The inverse causal direction, as shown in equation 8, is demonstrated by equation 9. The error term in the Granger causality test is assumed to be autocorrelation-free.

In contrast, the maximum Eigenvalue and trace tests are combined in the Johansen cointegration test. Using the maximum Eigenvalue test, Johansen's alternative of $r+1$ cointegrating equations is sharply contrasted with the null hypothesis of r cointegrating equations (Søren, 1991). The following formulas can be used to express the logarithm of maximum eigenvalues and trace statistics:

$$LR_{Trace}(r/n+1) = -T \sum_{i=r+1}^P \ln(1-\lambda) \quad (10)$$

$$LR_{Max}(r/n+1) = -T * \text{Log}(1-\lambda) \quad (11)$$

The sample size is denoted by T in this context, and the word "maximum eigenvalue" is key. Trace statistics are able to evaluate the antecedent of cointegrating equations (r) as well as the alternative hypothesis of n cointegrating equations. Consequently, n represents the number of variables that are being considered.

4. DATA ANALYSIS AND RESULTS

In this section of the study, the conclusions drawn from the data analysis carried out with the model equations given in the model specifications are dealt with. The findings include error correction models, multicollinearity tests, stationarity tests, cointegration tests, correlation matrices, summary statistics, and the Granger causality test. To ensure a comprehensive presentation of the results in the error correction models, significance levels of, ***1%, **5%, and *10% are interpreted as integral parts of the findings.

4.1 Summary Statistics

The summary statistics offer a brief and informative overview of the dataset utilized in the study, providing a quick, descriptive grasp of the data's values. This concise overview serves as a valuable tool for gaining a rapid insight into the dataset's characteristics, pinpointing trends and patterns. Table 5 contains summary statistics for each variable utilised in the inquiry, including the average, median, highest, lowest, and deviation from the mean.

Table 5: Summary Statistics

Variable	Mean	Median	S.D.	Min	Max
LGDP	3.37	3.31	0.534	2.58	4.25
Linflation rate	2.80	2.74	0.601	1.59	4.09
LCapital expenditure	8.07	8.30	2.58	3.27	11.7
LRevenue expenditure	7.60	7.81	2.49	3.09	11.0
LTotal tax revenue	7.66	7.57	2.15	4.20	11.1

Source: Author's calculations using Gretl

As presented in table 5, the average annual economic growth rate of 3.37 percent reveals Ghana's GDP has been growing steadily and positively over the research period. This suggests a generally healthy economic environment characterized by consistent growth.

The relatively low standard deviation of 0.534 percent implies that GDP values are closely clustered around the mean. This points to a certain level of stability in economic growth, indicating that the economy has not experienced drastic fluctuations. The range of GDP values, represented by the minimum and maximum values, provides insights into the lowest and highest points of economic activity during the study period, offering a perspective on the

overall economic variability. On the contrary, the variation of 0.6 percent in inflation rates indicates fluctuations over the research period, reflecting periods of both lower and higher inflation. This suggests that inflation dynamics were not uniform, and the economy experienced changes in price levels. An adequate level of general inflation is represented by the 2.8 percent mean inflation rate. This can be seen as a balance that avoids both deflationary pressures and excessive inflationary risks, contributing to economic stability. The range between the minimum and maximum inflation rates (1.59 percent to 4.09 percent) highlights the span of inflationary pressures experienced during the study period, providing a sense of the inflationary challenges faced by the economy.

Additionally, the relatively high standard deviation of 2.58 in capital expenditure indicates considerable variability, suggesting fluctuations in government investment over time. This variability may be due to shifts in capital project priorities or changes in government policy. The mean value of 8.07 for annual capital expenditure reflects the average rate of investment in capital projects by the Ghanaian government. The range of 3.27 percent to 11.7 percent, which is the minimum and maximum values, shows how varied government capital spending can be. Furthermore, government's annual revenue expenditure ranged from a minimum of 3.09 percent to a maximum of 11 percent. The average annual revenue expenditure is 7.6 percent, with a median of 7.81 percent and a variation rate of 2.49 percent. The variation in revenue expenditure suggests changing fiscal priorities or economic conditions influencing government spending. The middle figure reveals the core propensity of these expenses. Lastly, the average total tax revenue over the study period is 7.66 percent, with a variation rate of 2.15 percent. The total amount of tax income that was recorded ranged from 4.20 percent to 11.1 percent, and the median total tax revenue stood at 7.57 percent. The variation in total tax revenue indicates shifts in the government's ability to collect taxes, possibly influenced by economic conditions or changes in tax policies. The median and range offer additional context on the distribution of tax revenue values. In summary, the analysis provides a nuanced understanding of Ghana's economic performance, highlighting both positive trends and areas of variability.

Table 6: Correlation Matrix

LGDP	LInflation rate	LCapital expenditure	LRevenue expenditure	LTotax revenue	
1.000	-0.453	0.982	0.982	0.996	LGDP
	1.000	-0.444	-0.446	-0.453	LInflation rate
		1.000	0.999	0.988	LCapital expenditure
			1.000	0.989	LRevenue expenditure
				1.000	LTotax revenue

Source: Author's calculations using Gretl

The correlated scores in Table 6 show for years links throughout the variables. The association between GDP and inflation, in particular, is adverse, as expressed in the figure -0.453. In contrast, the coefficients of 0.982, 0.996, and 0.982 demonstrate that, over the long term, capital expenditure, total tax revenue, and revenue expenditure exhibit a positive relationship with Ghana's economic growth, respectively. The computation of correlation coefficients using data from 1990 to 2022, with a two-tailed 5% critical value, yields a value of 0.344. This suggests that, over the long term, the selected variables exhibit a significant positive relationship with Ghana's GDP.

4.2 Collinearity Diagnostic

The primary benefit of a collinearity diagnostic is to notify the statistician of a potential problem. This study employed a technique put forward by Belsley et al. (1980) to investigate the potential for multicollinearity between the variables. According to BKW's experiments, indicators falling between 0 and 10 suggest minor and relatively close dependencies, whereas indices between 10 and 30 indicate a higher degree of near dependencies. Table 7 presents the findings from the collinearity test

Table 7: BKW Collinearity Diagnostics

lambda	Cond	const	DLGDP	DLInfl~	DLCapi~	DLReve~	DLTota~
4.395	1.000	0.003	0.007	0.000	0.006	0.008	0.010
1.030	2.065	0.000	0.001	0.816	0.002	0.001	0.002
0.226	4.413	0.006	0.003	0.075	0.242	0.001	0.567
0.175	5.016	0.011	0.524	0.073	0.085	0.102	0.146
0.128	5.862	0.034	0.016	0.021	0.155	0.870	0.175
0.046	9.732	0.944	0.449	0.014	0.510	0.019	0.100

Source: Author's calculations using Gretl

According to the collinearity test conducted following the methodology of Belsley, Kuh, and Welsch (BKW), the results showed no significant interdependencies or perfect combinations among the variables. This conclusion is drawn as the highest Cond value observed is less than 10. Therefore, the analysis yields no evidence of multicollinearity.

4.3 Vector Autoregression (VAR) Lag-Length Selection

The performance of models needs an appropriate lag length to ensure optimization and efficiency in the outcome. In pursuit of efficient results for this study, a VAR lag order selection process was conducted to determine the most suitable lag to be employed. The three main information criteria used in this selection procedure were the Hannan-Quinn criterion, the Schwarz Bayesian criterion, and the Akaike criterion. The result of the vector autoregressive lag-length selection is shown in Table 8.

Table 8: VAR Lag Selection

lags	loglik	p(LR)	AIC	BIC	HQC
1	-920.819	None	63.387	64.789	63.836
2	-888.028	0.00002	62.868	65.437	63.690
3	-792.209	0.00000	58.147*	61.883*	59.342*

Source: Author's calculations using Gretl

The analysis of the VAR lag selection, as indicated by the p(LR), consistently pointed to lag 3 as the most suitable choice. As such, the third lag is the best decision for the model's calculations.

4.4 Stationarity Test

In econometric data analysis, it is vital to assess time series variables' attributes. The variables' stability or instability can be determined using these features. The Augmented Dickey-Fuller test was employed in this work to determine the possibility of unit root structures. Dickey and Fuller (1979) first introduced the ADF test which is used to determine if a time series has equal values or a unit root, indicating that the variable follows a random walk. When a time series has a unit root, it suggests that the data may have a tendency to walk randomly. The ADF test results are displayed in Table 9.

Table 9: Stationarity Test at Levels and First Difference

Variables	Sample size	ADF T-stat	p-value	Critical Value	ADF T-stat	p-value	Critical Value
LGDP	1992-2022	-2.260	0.455	-0.160	-3.623	0.005	-0.625
LInflation rate	1992-2022	-0.206	0.612	-0.008	-4.821	0.000	-2.863
LCapital expenditure	1992-2022	-0.947	0.949	-0.068	-6.595	0.000	-1.208
LRevenue expenditure	1992-2022	-2.721	0.070	-0.024	-6.272	0.000	-1.170
LTotax revenue	1992-2022	-1.802	0.704	-0.204	-6.202	0.000	-1.140

Source: Author's calculations using Gretl

The possibility of a unit root at the series level is evaluated using the ADF test's null hypothesis ($H_0: \rho = 0$). In contrast, it is suggested that there isn't a unit root by the alternative hypothesis ($H_1: \rho \neq 0$). In the event that the p-value surpasses the critical value, a unit root is detected at the 5% significance level. When the likelihood under the alternative hypothesis is less than five percent, it indicates the presence of a unit root. To determine the best options, the ADF test is conducted with and without constants and with or without trends (time). In accordance to the ADF tests, all variables are having unit root at their levels, but they became stationary at their first differencing.

Figures 2 and 3 depict the variables at their original levels and in their first difference form

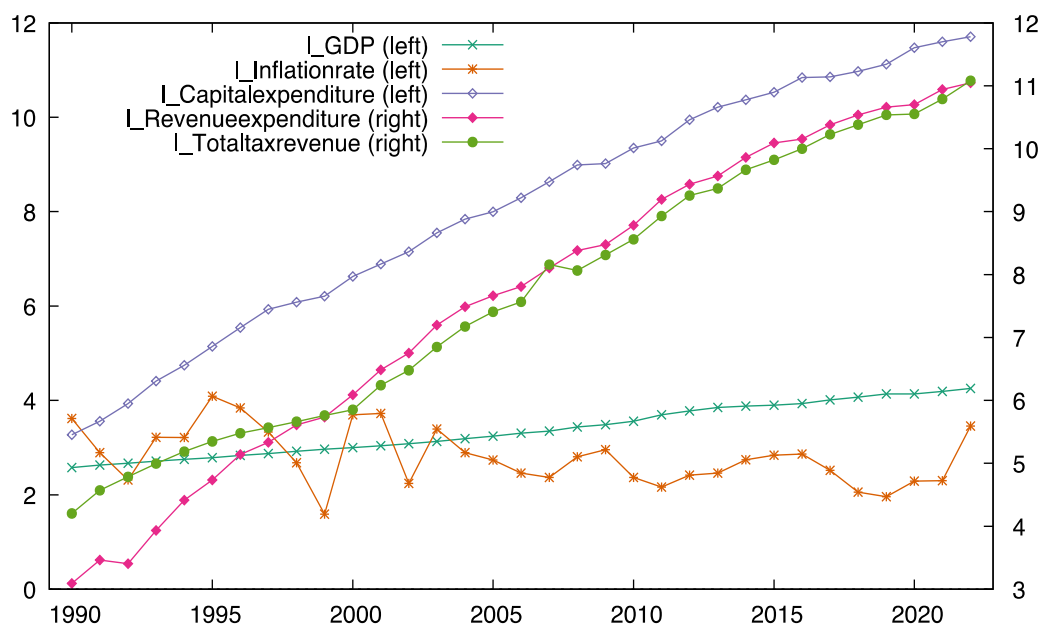


Figure 2: Time series plot at level

Source: Author's computation using Gretl

Figure 2 indicated that variables followed the random walk property which means that the series are non-stationary at level.

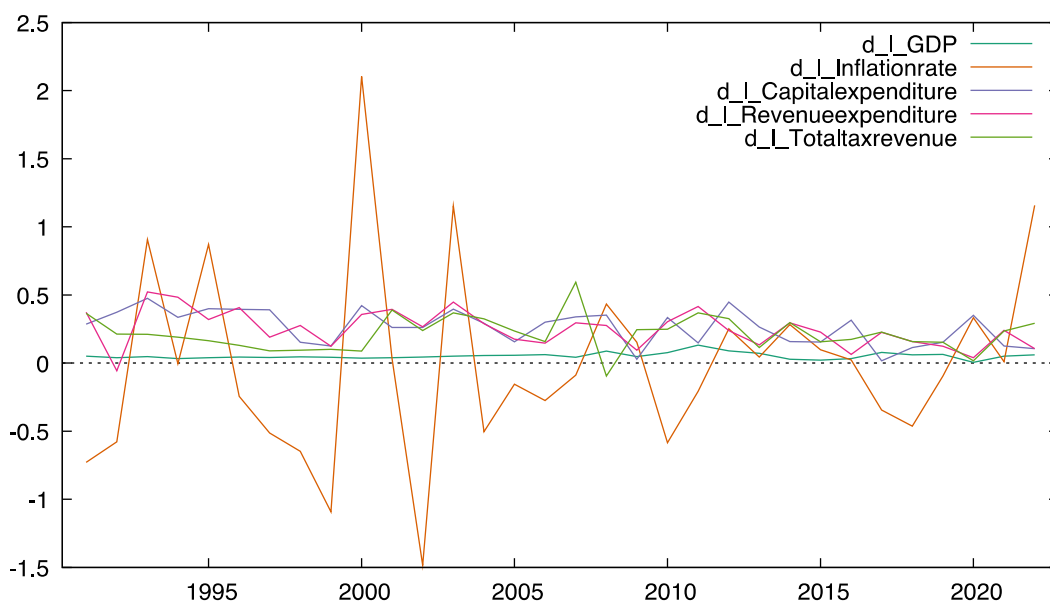


Figure 3: Time series plot at first difference

Source: Author's computations using Gretl

Figure 3 illustrates the white noise property by displaying the initial difference of the series. The variables therefore show integration at order I (1).

4.5 Johansen Cointegration Test

Assessing the long-term correlations between the selected variables became essential when it was discovered that they were integrated at the first-order difference. The prolonged underlying connections between factors relevant to Ghana's economic growth were determined using the cointegration test. The trace test and the maximum likelihood test are the two primary tests that are performed with the Johansen test framework. Ranking 0 and 4 are the primary ranks, and the number of ranks dictated the cointegration equation. There is no collaboration between the series, according to the preliminary point of view, which is related to rank 0. According to the second null hypothesis, which corresponds to rank 1, there may only be one cointegration equation between the variables at most. The trace and likelihood test p-values determine whether these null hypotheses are accepted or rejected. To be more specific, the probability values from both tests must be beyond the critical values at a level of significance equal to five percent in order to determine the presence of the interaction. The null point of view, on the other hand, is disproven if the probability value is less than the critical value. The interdependence equation is obtained from the renormalized beta values in the Johansen test output, with the coefficients of the independent variables having their signs inverted. With GDP as the explained factor, the Johansen test results are shown in Table 10.

Table 10: Johansen linking GDP and explicable variables

Rank	Eigenvalue	Trace test	P-value	Likelihood maximum test	P-value	Conclusion
0	0.994	239.78	0.000	156.27	0.000	Rejected
1	0.711	83.507	0.000	37.294	0.001	Rejected
2	0.627	46.213	0.000	29.603	0.002	Rejected
3	0.399	16.609	0.032	15.292	0.032	Rejected
4	0.042	1.317	0.251	1.317	0.251	Not Rejected

Sample period: 1992-2022; Source: Author's calculations

The outcomes derived from the Johansen test, detailed in Table 10, demonstrate the presence of cointegration among the variables, proving an unrelenting correlation with economic growth in the overall Ghanaian economy. The determination of cointegration primarily relied on both the trace and likelihood maximum tests. The findings reveal that up to Rank 3, there exist interconnecting links within the system, which is confirmed by the dismissal of the null idea for these ranks in both the Trace test and the Likelihood maximum test. This suggests that certain sets of parameters evolve concurrently in the long-run, indicating an unaltered balance connection among them. However, at Rank 4, the test refuses to disprove the null prediction, revealing that including another interrelated connection beyond Rank 3 fails to appreciably strengthen the model's descriptive ability. Thus, the Johansen Cointegration test highlights the significance of considering up to Rank 3 in the analysis, as it captures the fundamental lasting connection among the parameters under examination. The cointegration equation with GDP as the variable of interest, derived from the renormalized beta values, can be presented as follows:

$$\text{GDP} = -0.122(\text{inflation}) + 0.205(\text{capital expenditure}) - 0.091(\text{revenue expenditure}) + 0.020(\text{total tax revenue}) - 163$$

The cointegration equation in the long term indicates that Ghana's economic growth is adversely affected by both a 12% inflation rate and a 9% revenue expenditure. This implies that a reduction of one unit in both inflation and revenue expenditure would lead to an expansion of approximately twelve and nine percent, respectively, in Ghana's economic progress. Conversely, the positive coefficients for capital expenditure and tax revenue demonstrate their direct association with Ghana's GDP. Specifically, a 1% increase in both capital expenditure and tax revenue would correspond to a roughly twenty percent increase in the Ghanaian economy. This suggests that higher levels of capital expenditures and tax revenues serve to stimulate the socioeconomic development of Ghana.

4.6 Vector Error Correction Model

The VECM is a useful tool for determining how quickly the system returns to equilibrium after a deviation in any of the variables particular within the context of economic unpredictability and unexpected shocks. According to the VECM criteria, the error correction term coefficient must be opposing and significance at the level of five per cent in order to confirm the assumption of nullity. A high ECT coefficient, on the other hand, denotes the absence of long-term interdependencies between the variables and so fundamentally changes how the results

are interpreted. The findings from Table 11 specifically pertain to the VECM analysis with GDP as the dependent variable.

Table 11: VECM between GDP and explicating variables

Variable	Coefficient	Std. Error	t-ratio	p-value
Constant	0.048	0.018	2.649	0.014**
Δ LGDP_1	0.347	0.162	2.140	0.043**
Δ Inflationrate_1	-0.005	0.006	-0.852	0.402
Δ LCapital expenditure_1	0.013	0.034	0.414	0.682
Δ LRevenue expenditure_1	0.020	0.040	0.509	0.615
Δ Total tax revenue_1	0.079	0.031	2.595	0.016**
EC1	-0.043	0.018	-2.426	0.023**

Mean dependent var	0.052	S.D. dependent var	0.023
Sum squared residuals	0.009	S.E. of regression	0.019
R-squared	0.430	Adjusted R-squared	0.288
rho	0.124	Durbin-Watson	1.748

Sample Period: 1992-2022; Source: Author's calculations using Gretl

The constant term coefficient of 0.048 suggests that, when all lagged changes in the independent variables, GDP is expected to change by 0.048 units. This coefficient is analytically relevant at the 5% level, revealing its importance in describing fluctuations in GDP. A one-unit boost in the lagged change in log GDP corresponds to a 0.347-unit spike in GDP. This coefficient is highly considerable at the 5% level, demonstrating that modifications to GDP have a major effect on its future values. The statistic for the lagged shift in log inflation rate is -0.005. Although negative, this coefficient is not statistically significant at conventional levels ($p > 0.05$), suggesting that changes in inflation rates do not have a significant impact on GDP. The coefficients for the lagged changes in log capital and log revenue expenditure are 0.013 and 0.020, respectively. Neither indicator is of statistical importance at ordinary levels, meaning that adjustments in these parameters have no notable bearing on GDP. In contrast, the

index for the lagged change in log total tax revenue stands at 0.079. This suggests that a one-unit increase in the lagged change in total tax revenue is associated with a 0.079-unit increase in GDP. This coefficient is statistically significant at the 5% level, highlighting the significant impact of changes in total tax revenue on GDP. Lastly, the coefficient for the error correction term (EC1) is -0.043. This implies that for each period, the model adjusts towards equilibrium by approximately 4.3% of the deviation from equilibrium in the previous period. This component is considered relevant at the 5% level, exhibiting that the error correction mechanism plays an important role in explaining departures from long-term GDP equilibrium. As a result, the analysis regarding the inflation rate as the dependent variable can be found in Table 12.

Table 12: VECM between Inflation and the explicable variables

Variable	Coefficient	Std. Error	t-ratio	p-value
Constant	0.631	0.676	0.933	0.359
Δ LGDP_1	0.156	5.991	0.026	0.979
Δ LInflationrate_1	-0.270	0.229	-1.175	0.251
Δ LCapital expenditure_1	0.042	1.240	0.034	0.973
Δ LRevenue expenditure_1	0.202	1.487	0.136	0.892
Δ LTotal tax revenue_1	-0.249	1.127	-0.221	0.826
EC1	-0.679	0.648	-1.048	0.305

Mean dependent var	0.018	S.D. dependent var	0.700
Sum squared residual	12.867	S.E. of regression	0.732
R-squared	0.125	Adjusted R-squared	-0.093
rho	-0.089	Durbin-Watson	2.094

Sample Period: 1992-2022; Source: Author's calculations using Gretl

The resulting ECT coefficient in the VECM analysis using inflation rate as the dependent variable is negative, but it is not statistically significant. This result does not support the hypothesis that the variables are long-term dependent on one another. Table 13 shows the analysis with capital expenditure as the variable of concern.

Table 13: VECM between Capital expenditure and the explicable variables

Variable	Coefficient	Std. Error	t-ratio	p-value
Constant	−0.047	0.105	−0.445	0.659
Δ LGDP_1	−0.126	0.934	−0.134	0.893
Δ Inflationrate_1	−0.013	0.035	−0.369	0.714
Δ Capital expenditure_1	0.259	0.193	1.341	0.192
Δ Revenue expenditure_1	−0.287	0.232	−1.240	0.227
Δ Total tax revenue_1	0.212	0.175	1.206	0.239
EC1	0.289	0.101	2.863	0.008***

Mean dependent var	0.262	S.D. dependent var	0.128
Sum squared residual	0.313	S.E. of regression	0.114
R-squared	0.367	Adjusted R-squared	0.209
rho	−0.047	Durbin-Watson	2.035

Sample Period: 1992-2022; Source: Author's calculations using Gretl

In the VECM analysis with capital expenditure as the explained variable, the observed error correction term coefficient is positively significant. Nevertheless, the presence of a positive ECT coefficient changes the interpretation of the results and fails to confirm a long-term relationship among the variables. Conversely, you can find the VECM analysis with revenue expenditure as the explained variable in Table 14

Table 14: VECM between Revenue expenditure and the explicable variables

Variable	Coefficient	Std. Error	t-ratio	p-value
Constant	0.265	0.1233	2.155	0.041**
Δ LGDP_1	-0.703	1.092	-0.643	0.525
Δ Inflationrate_1	-0.013	0.041	-0.318	0.752
Δ Capital expenditure_1	0.402	0.226	1.778	0.088*
Δ Revenue expenditure_1	0.106	0.271	0.3914	0.698
Δ Total tax revenue_1	0.111	0.205	0.540	0.593
EC1	-0.151	0.118	-1.285	0.211

Mean dependent var	0.244	S.D. dependent var	0.136
Sum squared residual	0.427	S.E. of regression	0.133
R-squared	0.236	Adjusted R-squared	0.045
rho	0.083	Durbin-Watson	1.679

Sample Period: 1992-2022; Source: Author's calculations using Gretl

The VECM analysis displays a negative ECT coefficient that is not statistically significant, with revenue expenditure acting as the dependent variable. As consequently, any continuous association between the indicators is ruled out. On the other hand, the lagged coefficient of capital expenditure has an upward effect on current revenue, which is highly probable at a level of 10 percent. Table 15 uses total tax revenue as the explication factor.

Table 15: VECM between Tax revenue and the explicable factors

Variable	Coefficient	Std. Error	t-ratio	p-value
Constant	0.155	0.126	1.226	0.232
Δ LGDP_1	1.314	1.119	1.175	0.251
Δ Inflationrate_1	0.023	0.042	0.542	0.592
Δ Capital expenditure_1	0.080	0.231	0.346	0.732
Δ Revenue expenditure_1	-0.039	0.278	-0.142	0.888
Δ Total tax revenue_1	-0.157	0.210	-0.749	0.461
EC1	0.010	0.121	0.083	0.934

Mean dependent var	0.210	S.D. dependent var	0.128
Sum squared residual	0.449	S.E. of regression	0.136
R-squared	0.088	Adjusted R-squared	-0.139
rho	0.029	Durbin-Watson	1.917

Sample Period: 1992-2022; Source: Author's calculations using Gretl

A positive ECT coefficient is shown by the VECM analysis results when total tax income is taken into account as the dependent variable; however, this coefficient is not statistically significant. In the case where the ECT coefficient is positive, there might not be any sustained interdependence among the variables. Therefore, when total tax revenue is used as the explained variable, the cointegration test findings are not supported. The results of the Johansen cointegration test, on the other hand, are consistent with the general conclusions from the VECM analysis using GDP as the target variable, indicating a persistent linkage of the variables affecting socioeconomic advancement.

4.7 Granger Causality Test

The Granger causality test is primarily concerned with evaluating the immediate links between parameters. It is commonly known that previous events can have an impact on present events, and to ascertain if a given variable can be utilised as a predictive of another variable, the Granger causality test is performed. A statistically significant F-statistics p-value at the 5% level is required to reject the null hypothesis in the Granger causality test. The presence of a transient linkage between the variables is the null hypothesis. It's also critical to understand that interactions between the series could be both unidirectional and bidirectional. A unidirectional relationship shows that the impact only flows in one direction, but a bidirectional relationship suggests that two variables might function as predicting factors for one another.

Table 16: Granger causality test of GDP and the explicable variables

Hypothesis	F-statistics	P-value	Decision
Linflation rate Granger cause GDP	0.563	0.578	Fail to reject
LGDP Granger cause inflation rate	0.696	0.509	Fail to reject
Lcapital expenditure Granger cause GDP	2.399	0.116	Fail to reject
LGDP Granger cause capital expenditure	0.011	0.988	Fail to reject
Lrevenue expenditure Granger cause GDP	1.621	0.222	Fail to reject
LGDP Granger cause revenue expenditure	0.044	0.956	Fail to Reject
Ltotal tax revenue Granger cause GDP	3.905	0.037	Rejected
LGDP Granger cause total tax revenue	0.375	0.691	Fail to reject

Sample Period: 1992-2022; Source: Author's calculations using Gretl

The findings suggest that fluctuations in the inflation rate do not Granger-cause changes in GDP, implying that shifts in inflation levels do not significantly affect economic output. Likewise, changes in GDP do not Granger-cause alterations in the inflation rate, indicating that economic performance does not notably influence inflationary pressures. Similarly, variations in capital expenditure do not Granger-cause changes in GDP, suggesting that shifts in

investment spending do not immediately impact economic output. In contrast, changes in GDP Granger-cause changes in total tax revenue, suggesting that shifts in economic activity directly affect tax revenues, likely mirroring the periodic pattern of tax receipts in response to GDP changes. This suggests the presence of a unidirectional short-term relationship, with total tax revenue influencing GDP in Ghana. Regarding revenue expenditure, the results indicate that changes in this variable do not Granger-cause changes in GDP, signifying that alterations in government spending patterns do not have an immediate significant impact on economic output. Similarly, changes in GDP do not Granger-cause changes in revenue expenditure, implying that economic performance does not prompt immediate adjustments in government spending patterns. Finally, changes in GDP do not Granger-cause changes in total tax revenue, suggesting that economic fluctuations do not directly lead to shifts in tax receipts.

4.8 Diagnostic Test

The primary purpose of conducting a normality test in this research is to evaluate whether the dataset adheres to a normal distribution or, more generally, to figure out whether it is feasible to assume that the data in question are normal. To ensure the correctness of the Vector Error Correction Model specification for the analyses, the Doornik-Hansen normality test was employed. This test furnishes the necessary confidence in the model's suitability for investigating the relationship between GDP and the explanatory variables, thereby enabling the study to draw meaningful conclusions. Regarding the normalcy test, the null hypothesis ($H_0: \rho = 0$) conventionally asserts that the data conforms to a normal distribution, while the alternative hypothesis ($H_1: \rho \neq 0$) suggests that the data deviate from normality.

Table 17: Doornik-Hansen normality test between GDP and the explanatory variables

Chi-square (10)	15.8043
P-Value	0.1054

Source: Author's calculations using Gretl

At the 5% significance level used in this study, the diagnostic test statistics shown in Table 17 failed to exhibit any signs of non-normality. As a result, the null concept was flatly invalidated.

4.9 Summary of Results

In summary, the evidence revealed by BKW collinearity diagnosis of Cond values below 10 affirms the variables autonomy and uniqueness. This obviously suggest that the variables in this model are not strongly correlated with each other, which has the tendency to provide accurate outcomes and predictions. To examined the stationarity status of the variables, Augmented Dickey Fuller Test showed that all the variables had unit root at their levels and become stationery after first differencing. The cointegration analysis reveals that over the long term, inflation rate and revenue expenditure exert a negative impact on economic growth in Ghana whereas capital expenditure and tax revenue have a positive influence on the country's progress. Additionally, the VECM which depicts a mechanism for adjusting deviations from the long-term equilibrium in the GDP revealed a roughly 4.3% speed of adjustment. This clearly shows the rate at which the GDP return to its equilibrium state following a shock or disturbance. The results of the Granger causality test revealed the absence of short-term relationships in both directions among GDP, inflation rate, capital expenditure, and revenue expenditure. This suggests that these variables cannot be employed to predict one another in the short term. However, the findings do indicate that total tax revenue has a short-term impact on economic growth and can be utilized as a predictive factor for Ghana's GDP. Consequently, this implies the existence of a unidirectional short-term relationship, with total tax revenue influencing GDP in Ghana.

5. DISCUSSIONS

The intent of this thesis is to explore the link between taxation, capital and revenue expenditures with regards to Ghana's economic growth. This objective is pivotal in addressing the research inquiries posed, contingent upon the results generated from the estimated models, which include the following questions:

5.1 What causal connection exists between Ghana's economic growth and its taxation system?

Theoretically and practically, there is a complex link between taxes and economic growth. The intricacy stems from the fundamental characteristics of taxation. An excessive tax rate can skew the supply and demand sides of the economy, even while a particular tax rate may be required to raise money for projects that promote growth. As a result, there are differences between theoretical models and actual data regarding the precise nature of the relationship between taxes and economic growth. Certainly, amid all this complexity, one undeniable fact emerges: taxes exert an influence on economic growth. This study inquires into the hyperlink involving tax income and Ghana's economic advancement. A unique feature of this study is its emphasis on investigating causality rather than solely assessing the impact. At a significance level of 3.7 percent, the null hypothesis asserting that total tax revenue does not Granger-cause real GDP is invalidated by the Granger causality test, this demonstrates an alliance of GDP and overall tax revenue. Instead, it accepts the alternative hypothesis, indicating that total tax revenue Granger-causes real GDP. We are unable to toss out the null proposition, which states that true gross domestic product does not Granger causes total tax revenue. As a result, a one-to-one connection from tax revenue and economic growth originates from total tax revenue to Ghana's economic advancement.

The discovery further reinforces the established understanding of the significant role of tax revenue in economic growth, as highlighted in prior studies by Stoilova (2017), Mullen and Williams (1994), and Karran (1985). The findings align with theoretical expectations, as government expenditures are predominantly funded through taxation. Effective allocation of such funds to productive infrastructure, the renovation of existing facilities, and investment in research and development can undoubtedly catalyze economic growth. This emphasizes the idea that tax revenue contributes to economic expansion when directed toward government-led capital projects and infrastructure development, leading to increased productivity, as argued by Nasiru (2012). Moreover, the derived coefficient of the error correction term, which is -0.043,

suggests an annual adjustment rate of almost 4.3 percent. The important and detrimental score confirms that the factors are interactive. The magnitude of the error correction term ratio shows that the preceding year's shocks collide to a permanent balance at a rate of roughly 4.3% of the total in the current year. A number of -0.122, on the other hand, implies that the inflation rate considerably impedes economic progress. Consequently, a one percent rise in inflation is associated with a substantial 12.2 percent reduction in economic growth. This underscores the notion that elevated inflation rates signify economic disruption. These results align with theoretical predictions, as articulated by Fischer in 1983, that inflation has detrimental long-term effects on economic growth.

5.2 How does Ghana's economic growth relate to Capital Expenditures?

The cointegration test revealed a prolonged consistent relatedness between capital expenditure and the real GDP. The research conducted by Appiah (2014) and Appiah-Konadu et al., (2016), in Ghana provided a confirmation to the existence of this positive long-term relationship. The positive coefficient validates the perspective put forth by the Keynesian growth model, which promotes the utilization of government expenditure as an investment to stimulate production and subsequently enhance economic growth (Ackah et al., 2014). Conversely, Adu and Ackah (2015) came to the conclusion that capital expenditure significantly hinders Ghana's economic expansion. Furthermore, there was no long-term correlation between capital spending and real GDP, according to the VECM. The lack of short-term interdependency between GDP and capital expenditure indicates that changes in capital expenditure do not immediately and directly affect GDP. For example, it takes time to invest in infrastructure such as constructing roads, schools, hospitals, and other physical assets. Therefore, the effects of this capital investment on real GDP are expected to materialize gradually and persist over an extended period, as indicated by Baltagi (2008).

5.3 Does Revenue Expenditure influence economic growth in Ghana?

The cointegration equation unveils a significant relationship, highlighting that revenue expenditure exerts a detrimental impact on the real GDP of Ghana. This phenomenon can be attributed to the predominance of Ghana's revenue expenditure, primarily driven by elevated wages and salaries following the implementation of the Single Spine Salary Scheme (Fair Wages and Salaries Commission, 2013). This discovery echoes the conclusion drawn by Mitchell (2005), suggesting that government spending imposes a behavioural subsidy cost on the economy. As the government continues to escalate its spending on various welfare programs, coupled with the increasing share allocated to wages and salaries within the government's revenue expenditure, it incentivizes counterproductive choices among economic agents. Consequently, individuals tend to opt for leisure over work, resulting in a decrease in real GDP. Nevertheless, these outcomes diverge from the findings of Ojonugwa et al. (2007), who identified a positive and statistically significant association between recurrent expenditure and economic growth, both in the long and short run, in Nigeria.

6. CONCLUSION AND RECOMMENDATIONS

This chapter focuses on the summary of the findings, including its policy implications and recommended actions Based on the findings about the role of tax policy on Ghana's growth in economy. It also addresses research limitations and makes hints for prospective research.

6.1 Conclusion

This thesis aims to assess the causal influence of taxes and the relationship of capital and revenue expenditures on the economic growth of Ghana. Achieving this objective necessitated the execution of various econometric data analyses. The employed methods encompassed multivariate time series techniques, including VAR lag selection, the Johansen cointegration test, the Vector Error Correction Model, the Granger causality test, and the unit root test.

This study used annual time series data from 1990 to 2022 from the Ghana Revenue Authority (GRA), the Ministry of Finance and Economic Planning, and the World Development Indicators (WDI). All of the series were specified to be non-stationary by the initial ADF unit root test on the selected variables. The series nevertheless remained stable after the variables' first difference, illustrating that they were integrated at the first order $I(1)$. Furthermore, when examining multicollinearity among the variables, the results indicated that none of the variables were perfect explanatory factors or combinations of others.

However, the cointegration test results indicated the existence of a long-term relationship between total taxes (comprising direct and indirect taxes), capital expenditure, revenue expenditure, and Ghana's economic growth. In particular, the test revealed a positive effect of both tax revenue and capital expenditure on Ghana's economic growth, whereas revenue expenditure exhibited a negative impact on economic growth. On the contrary, the results obtained from the vector error correction model suggested that taxes, capital expenditures, and revenue expenditures had noteworthy and affirmative influences on Ghana's economic growth. A unifacial unpremeditated liaison between total tax and economic growth was additionally established by the Granger causality test, which evaluated short-term impacts. This outcome is harmonious with other scientific forecasts and economic growth theory. Based on both short- and long-term data, the analysis concludes that taxes and economic growth in Ghana have a statistically significant positive causal link.

6.2 Recommendations

The study has clearly highlighted the fact that taxes have an essential effect on economic growth, which cannot be understated. In light of the aforementioned findings, it is advisable that the government intensify its efforts in revenue mobilization, considering that tax revenue plays a crucial role in funding government expenditures for developmental purposes. This is especially important given the affirmative bond of tax revenue on economic progress. To achieve this, the government should focus on expanding the tax base by including more economic activities and sectors as well as enhancing the efficiency of tax administration by bolstering and modernizing customs procedures and rationalizing tax exemptions.

Again, it is imperative for the government to prioritize expenditures that foster economic growth when embarking on national budgeting. A momentous allotment of public funds should be rumped into productive and capital sectors. Achieving this objective can be realized through the establishment of policy directives that outline the designated portions of public funds earmarked for capital investments. To guarantee effective implementation, robust monitoring measures should be put in place.

Finally, the government should give paramount importance to managing the public sector wage bill, which can be achieved by periodically reviewing wages in accordance with productivity enhancements. These comprehensive reviews will help curtail the substantial expenditure on salaries and wages for underperforming employees, ultimately fostering a more motivated workforce that can contribute to increased real GDP in Ghana through enhanced productivity.

6.3 Limitation of this study

The 32-year observation period from 1990 to 2022 was intended to provide an ample timeframe for assessing the success and efficacy of the analyses. It is imperative, therefore, to recognise certain drawbacks, such as the possibility of missing variables and problems with degrees of freedom that affected the choice of VAR-lag in order to optimise the many models this thesis presents. Factors that ought to have been incorporated into the models but weren't are known as omitted variables, and they might skew the outcomes. Moreover, during the observation period, there were structural disruptions such as economic crises or changes in the nation's economy, which might have an effect on the long-term equilibrium. Since 1990, numerous significant events have shaped the economic landscape, including the 2008 financial crisis and the recent Covid-19 pandemic.

6.4 Future Research

This study has significantly enhanced our comprehension of the linkage of tax policy on Ghana's economic growth. Future research endeavours could further enrich our insights by using different variables which contributes to tax policy effects on economic growth in Ghana as well as expanding the sample size for an inadept analysis.

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8. List of Figures

Figure 1: Chart flow of Methodology	36
Figure 2: Time series plot at level.....	47
Figure 3: Time series plot at first difference.....	47

9. List of Tables

Table 1: Categorization of taxes in Ghana	9
Table 2: PIT Monthly tax rate in Ghana-Effective 1st May, 2023.....	10
Table 3: PIT Annual tax rate in Ghana-Effective 1st May, 2023:.....	10
Table 4: Overview of the variables, probable signs of their coefficient and data origin	37
Table 5:Summary Statistics	42
Table 6: Correlation Matrix	44
Table 7: BKW Collinearity Diagnostics.....	45
Table 8: VAR Lag Selection	45
Table 9: Stationarity Test at Levels and First Difference	46
Table 10: Johansen linking GDP and explicable variables.....	48
Table 11: VECM between GDP and explicating variables.....	50
Table 12: VECM between Inflation and the explicable variables	51
Table 13: VECM between Capital expenditure and the explicable variables	52
Table 14: VECM between Revenue expenditure and the explicable variables	53
Table 15: VECM between Tax revenue and the explicable factors	54
Table 16: Granger causality test of GDP and the explicable variables	55
Table 17: Doornik-Hansen normality test between GDP and the explanatory variables	56